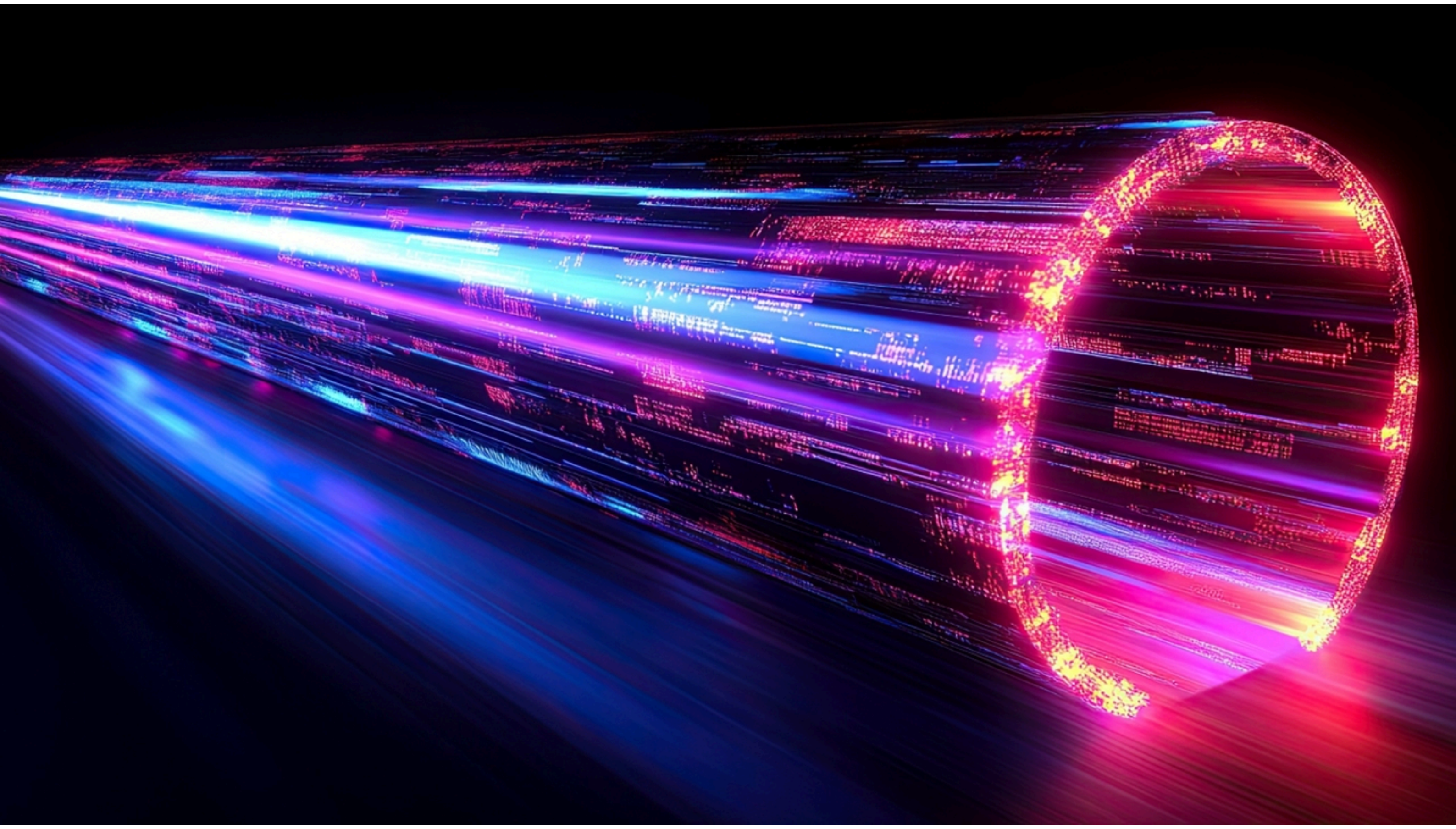
**Macquarie Equity Research**
13 November 2025

Macquarie Ecosystem Research

# AI networking

## Explosive market growth ahead for enablers



## Key Points

- Hyperscalers are now prioritising network expansion in their AI ramp ups to meet computing demand due to the physical limitations of silicon upgrades.
- We see high-speed interconnect enablers for AI networking, such as switches and transceivers, as key beneficiaries. We expect these two markets to deliver 50% and 70% revenue CAGRs over the next three years.
- We explore the entire ecosystem and highlight the biggest winners: Innolight, LandMark, VPEC and ISU Petasys.

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For important disclosures and analyst certification, refer to page 29 or go to www.macquarie.com/research/disclosures.

# Contents



# Associated initiations

Innolight (300308 CH; Outperform)

LandMark (3081 TT; Outperform)

VPEC (2455 TT; Outperform)

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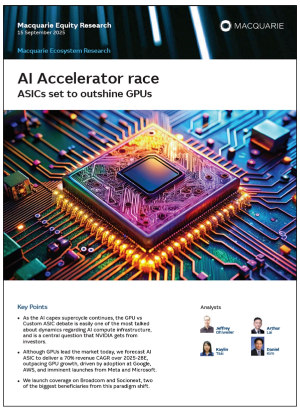
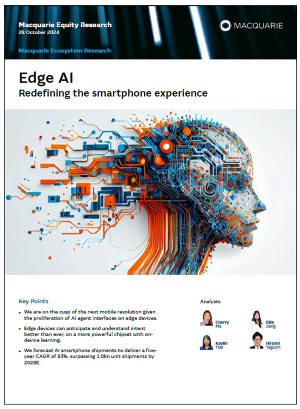
# More Macquarie Ecosystem Research





# Executive summary

Macquarie's Technology Research Team has been exploring subsegments that will deliver outsized & secular growth in the AI super-cycle we are in, highlighted in our reports on Beyond Moore's Law, Semiconductor IP, and Asics vs GPU. We now turn our attention to **AI Networking.**

Network expansion will start to take higher priority in hyperscalers' AI ramp-up plans.

**Network upgrades crucial for hyperscalers' AI dreams**

AI networking is fundamentally different from traditional data-centre networking architecture. Given the physical limitations of AI accelerators, in contrast to surging AI workloads, we believe network expansion will start to take higher priority in hyperscalers' AI ramp-up plans.

AI networking is the integration of AI and machine-learning technologies into networking systems to improve network intelligence, performance and security, and support AI workload at scale. AI networking is a critical step for improved AI training performance and (expensive) graphics-processing-unit (GPU) utilisation, as it can connect AI accelerators in scale (thousands or millions) to then act as a huge single supercomputer, or a "cluster" which is the backbone of data-centre infrastructure. Larger clusters can reduce training time significantly. The lead time for each generation of model will determine the winners, and therefore AI companies are racing to build larger and larger clusters.

Meta, for example, has been growing its "cluster" size by 4x every year since 2023, while Broadcom has indicated that most of its top customers are aiming for a 1mn + XPU (any processing units) cluster by end-2027.

Core components: Interconnects, High performance switches, and Compute accelerators (XPUs).

**Core components of AI networking are:**

- **Interconnects:** AI networks connect thousands of compute accelerators - including graphics processing units (GPUs) and data processing units (DPUs) - using copper (in case of Scale Up) or optical links, cabling and transceivers (in case of Scale Out) optimised for high-speed, lossless data movement at scale. Interconnects form the backbone of digital communication, linking data and services across disparate systems, data centres, clouds and organisational boundaries.
- **High performance switches**: AI networks use advanced hardware (such as 800G and 1.6T Ethernet or InfiniBand) and optimised controllers for ultra-fast, low-latency data exchange between compute nodes, data storage and orchestrator platforms. Switches often feature specialised packet processors and deep-packet buffers to accommodate spikes in AI traffic and prevent packet loss.
- **Compute accelerators (XPUs)**: AI networks rely on powerful processors (DPUs, GPUs and other AI-specific processors), organised in large, interconnected clusters, to implement parallel processing and accelerate AI model training and inferencing.

We are bullish on network transceivers and switches, and related component makers.

Against this backdrop, **we are particularly bullish on network transceivers and switches, and related component makers**, covering laser, PCB, and copper clad laminate (CCL). We expect the AI network hardware market to outgrow the Compute Accelerator(XPU) market , given its early stage of clustering with a lower base. As premium network hardware replaces low-mid spec hardware, we expect network hardware to enjoy secular growth or even face a shortage in the future. We see capacity constraints and longer ramp-up cycles in the upstream supply chain - such as epi-wafer for laser chips, copper foil and glass fibre for PCB - resulting in a near-term supply shortfall. ***We estimate a 50% revenue CAGR for the AI networking switch market over 2024-27E, and a 70% revenue CAGR for transceivers vs 34% for XPU shipments over the same period.***

**Roadmap for AI network evolution**

**Scale Out vs Scale Up**: Larger clusters are enabled by additional segments of scale up and scale out. In AI networking, "scale out" means adding more machines to distribute large workloads across a massive computing cluster (between-rack networks), while "scale up" means adding more resources to a single machine for ultra-low latency (in-rack fabric). Scale

Out is about horizontal growth by adding more servers, while Scale Up is about vertical growth, like increasing connected GPUs and memory size in one server rack.

Expect three waves of network component revenue growth

**We expect network component revenue to realise growth in three waves:** the first wave linked to the next-gen AI server rack ramp-up (ie, GB300 and Nvidia CX8 800G upgrade), the second being ASIC 800G network upgrade, and the third being upgrades in 2H26 with Nvidia's Rubin-based rack (this is Nvidia's next generation AI platform following Blackwell) ramp up and even larger ASIC clusters build-out. While positive on Co-Packaged Optics (CPO) - a technology that integrates optical components like transceivers with electrical components such as switches or processors, onto a single package substrate - we are conservative on the adoption timeline at this stage.

**Further ahead, we are positive on long-run demand for switches and transceivers**.

We highlight that the scale-up market could provide a larger total addressable market (TAM) for switches and transceivers once standardised. Yet at present, it remains a proprietary market; when NVIDIA designs Scale Up technology, they set the market rule and ship alongside their AI server, with only their vendors for the time being. Note that Nvidia (GPU) is using copper cable as its only solution for scale-up, but ASICs are exploring the possibility of using optical transceivers as their Scale-Up interconnect. As such, ASICs growth, as discussed in our recent AI Accelerator Race report, could potentially unlock an even larger scale-up market opportunity.

We expect the open-standard Scale-Up switch to ship in 2H26, at which point we may start to see the market open up to new suppliers, creating more opportunities. The advent of data-centre interconnect (DCI) for super clusters may start in 2027E; we expect more visibility going into 2026.

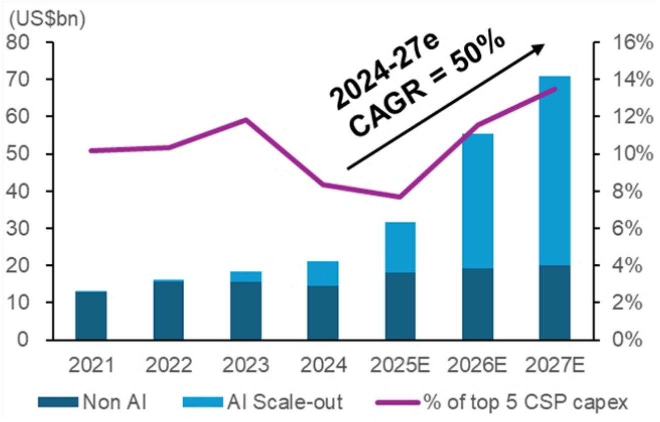
**We initiate coverage on the transceiver value chain**

We initiate on leading transceiver maker Innolight, and upstream names VPEC and LandMark, at Outperform. We also rate multi-layer PCB maker ISU Petasys Outperform.

Transceivers are one of the most important components enabling longer-distance/ higher-data rate transmissions. If we limit the power of the electrical signal, the signal can only travel a limited distance. At higher data rates, signal loss becomes more severe. We can convert electrical signals to optical signals with a reasonable conversion power cost, which enables much longer transmission distance, which is where transceivers come in, translating electrical signals to optical signals and vice versa.

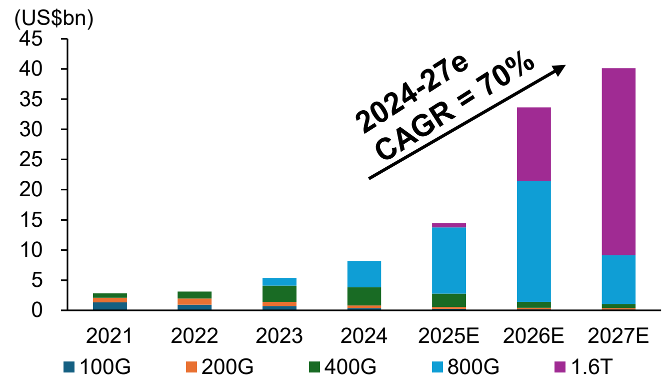
We initiate on three companies in the AI networking ecosystem with Outperform ratings: **Innolight** (300308 CH) a leading transceiver maker, and **VPEC** (2455 TT) and **LandMark** (3081 TT), upstream component makers which should enjoy the supply bottleneck). We expect the overall transceiver market to record a 70% revenue CAGR, to become a US$40bn market by 2028E. We are also positive on multi-layer PCB maker **ISU Petasys** (007660 KS), as a beneficiary of the same demand growth.

**Figure 1 - We project a 50% AI networking switch CAGR over 2024-27E**




*Source: Macquarie Research, Nov 2025*

**Figure 2 - . . . and a 70% transceiver market CAGR**




*Source: Macquarie Research, Nov 2025*

Stellar market growth expected for switchers and transceiver

# Ways to play the theme

We forecast a 70% transceiver market revenue CAGR over 2024-27E, and we anticipate rapid expansion of the switch market, estimating a 50% revenue CAGR over the same period. These growth rates are abnormal and non-cyclical**,** with supply bottlenecks likely to contribute. We have already seen shortages of key components such as copper clad laminate (CCL) and electro-absorption modulated lasers (EML), and we expect the supply crunch is only just becoming apparent, with more serious spillover effects yet to come. Specifically, we are reviewing upstream key component segments, as we see opportunities here for new-entrants.

**Silicon plays: GPU / ASIC / Switch Silicon / SerDes**

- **Broadcom (AVGO US; OP)** is the absolute king in the high-speed networking space, with leading 224G SerDes IP to benefit from cluster development and ASIC scale-up.
- **MediaTek (2454 TT; OP)** is a potential player in ASIC with 224G SerDes IP ready.
- **Alchip (3661 TT; OP)** is likely to return to strong growth in 2026 on its key AI ASIC customer's 3nm ramp. However, Alchip has also secured several networking-related orders that should also enter mass production in 2H26. The company partners with Synopsys (and others) on IP and has announced a partnership with Astera Labs for "AI Rack-Scale Connectivity".

**Hardware plays: PCB / CCL / Substrate / Switch System**

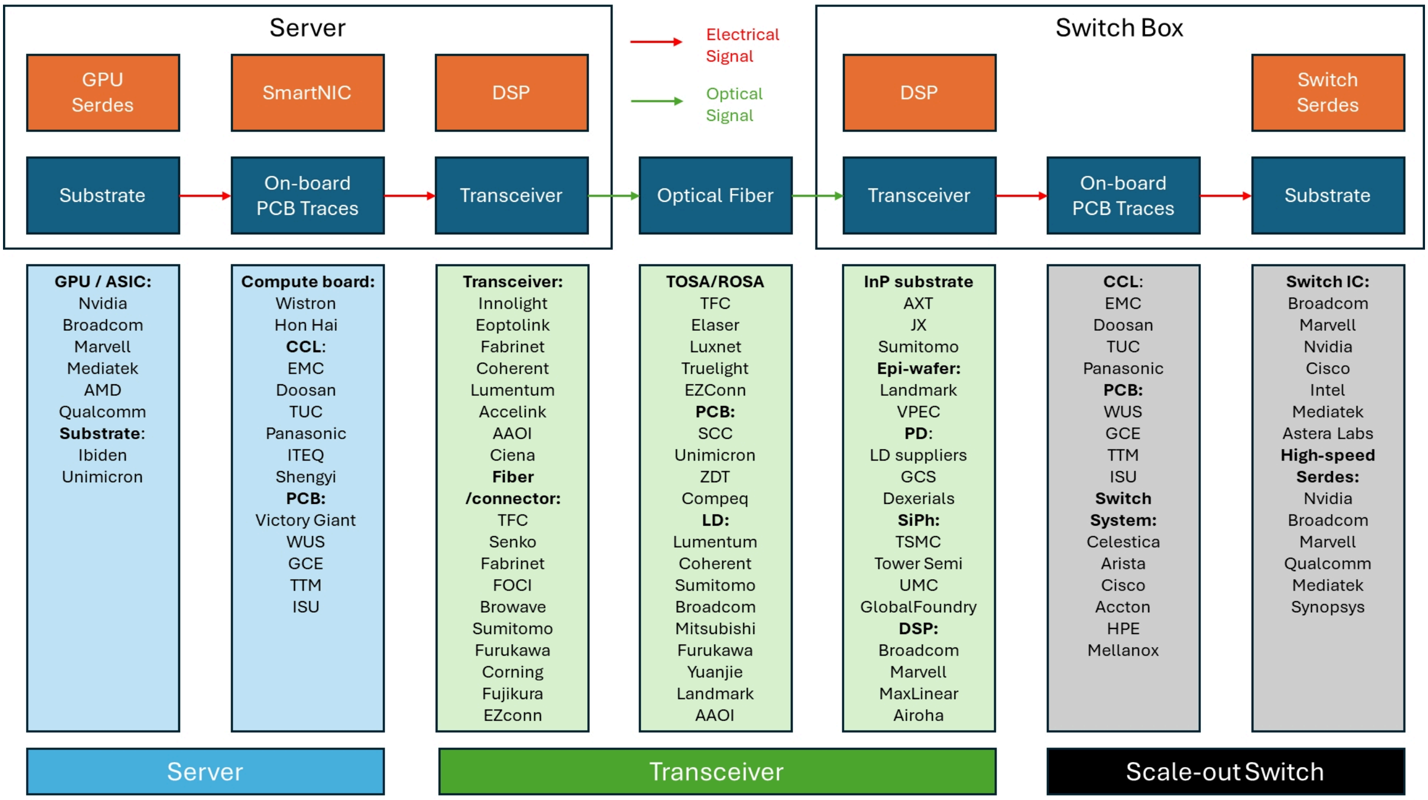
- **Unimicron (3037 TT; OP)**: Unimicron makes substrates for Nvidia's GPUs, ASICs and networking switches. It also makes HDI boards for optical modules. The company has noted a T-glass (glass fibre) supply shortage and sees no meaningful improvement until 2H26. We expect this to affect BT substrates more than ABF as the supply chain prioritises AI server-related supply. We estimate overall AI substrate/PCB revenue contribution to be 28%/30% of 2025/26E revenue.
- **ISU Petasys (007660 KS; OP)**: ISU is a pure data centre multi-layer board (MLB) play, with approximately 70% of total sales being AI-driven. ISU is the largest producer in the high-end (18+ layer) MLB market, especially with a strong presence in switch MLB production (for Google/Arista). Going forward, 800G network switches should require multi-stack MLBs (higher-end; 30+ layer) for their motherboards, and we believe this product has much higher entry barriers compared to the current industry mainstay 18+ layer high-end MLB. We are expecting AI scale-out network switch growth to be the biggest growth driver for ISU, underpinning a doubling in sales over the next three years. See full report here.
- **Elite Material (2383 TT; N)**: EMC supplies copper clad laminate (CCL, the raw material for printed circuit boards) to Nvidia and ASIC servers, as well as networking switches. We expect tight industry supply for high-end glass fibre and copper foil but note the 1-2-year lead time for EMC in expanding production capacity. We estimate total infrastructure (mostly AI server and networking) to contribute 69%/74% of 2025/26E revenue.
- **Accton (2345 TT; OP)**: Accton is one of the major beneficiaries of the networking trend, as it is the main white box switch supplier for Amazon and it has stated it expects to secure another hyperscaler customer in 2026. 800G revenue contribution is taking off in 2025 and we expect small shipments of 1.6T in late 2026. We believe the upsides for Accton are (1) adoption of scale-up networking switches that will significantly increase ASIC server's networking dollar content (ie. Accton's addressable market), and (2) share gain in Meta or other hyperscalers on higher white box adoption.

**Optical interconnect: transceiver assembly / laser chips / CPO**

- **Innolight (300308 CH; OP)** is the undisputed leader in the high-speed transceiver market.
- **VPEC (2455 TT; OP), LandMark (3081 TT; OP)**: Tight supply of laser chips, and their epiwafers served as a key upstream component for laser chip.
- **Chroma ATE (2360 TT; OP)**: We believe testing equipment will be the first-wave beneficiary of co-packaged optics (CPO), and Chroma is already developing burn-in equipment ready for CPO mass production testing.

**Key risks**: Delay in next-gen AI server or ASIC roadmap. Slower than expected switch / transceiver demand. Shortage in key components.

**Figure 3 - Scale-out Opportunity for Optical Communication (public companies)**




*Source: Macquarie Research, Nov 2025*

## Figure 4 - AI Networking - Valuation comparisons

| | | | Mkt Cap | CP | TP | TSR | EPS (LC) | | | PER (X) | | | P/BV (X) | | | ROE (%) | | | Div yield (%) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ticker | Company | Rating | (US$m) | (LC) | (LC) | (%) | 24A | 25E | 26E | 24A | 25E | 26E | 24A | 25E | 26E | 24A | 25E | 26E | 24A | 25E | 26E |
| **GPU ASIC/Switch IC/Serdes IP/DSP** | | | | | | | | | | | | | | | | | | | | | |
| NVDA US | Nvidia | NR | 4,709,340 | 193.8 | NA | NA | 1.2 | 2.9 | 4.5 | 167.9 | 67.6 | 43.0 | 115.8 | 62.2 | 33.3 | 85.3 | 116.9 | 92.6 | 0.0 | 0.0 | 0.0 |
| AVGO US | Broadcom | O | 1,650,136 | 349.4 | 420.00 | 21% | 2.9 | 6.8 | 9.5 | 121.2 | 51.6 | 36.7 | 25.3 | 21.7 | 15.0 | 51.8 | 45.4 | 48.3 | 0.3 | 0.6 | 0.7 |
| MRVL US | Marvell | NR | 77,011 | 89.3 | NA | NA | 1.5 | 1.5 | 2.7 | 59.2 | 58.6 | 32.6 | 5.2 | 5.7 | 5.3 | -5.7 | -6.3 | 5.3 | 0.3 | 0.3 | 0.3 |
| 2454 TT | MediaTek | O | 65,312 | 1,260.0 | 1,680.00 | 41% | 66.9 | 64.9 | 65.3 | 18.8 | 19.4 | 19.3 | 5.1 | 5.6 | 6.4 | 27.8 | 27.3 | 31.0 | 4.4 | 5.5 | 7.5 |
| INTC US | Intel | NR | 180,735 | 37.9 | NA | NA | 1.1 | -0.1 | 0.3 | 36.1 | n.a. | 110.8 | 1.5 | 1.7 | 1.6 | 1.6 | -18.3 | 0.7 | 2.0 | 1.0 | 0.0 |
| ALAB US | Astera Labs | NR | 26,651 | 157.8 | NA | NA | -0.7 | 0.8 | 1.8 | n.a. | 187.8 | 88.6 | na | 26.5 | 20.7 | N/A N/A | 3.0 | 21.5 | 0.0 | 0.0 | 0.0 |
| QCOM US | Qualcomm | NR | 189,214 | 176.7 | NA | NA | 8.8 | 10.9 | 12.0 | 20.1 | 16.3 | 14.7 | 8.6 | 7.8 | 8.4 | 38.0 | 37.6 | 29.8 | 1.8 | 1.9 | 2.0 |
| SNPS US | Synopsys | NR | 74,078 | 398.8 | NA | NA | 11.8 | 13.0 | 13.2 | 33.7 | 30.6 | 30.3 | 9.2 | 5.1 | 2.3 | 22.6 | 26.5 | 9.0 | 0.0 | 0.0 | na |
| 6526 TT | Airoha | NR | 2,511 | 469.0 | NA | NA | 6.5 | 16.1 | 18.0 | 72.5 | 29.1 | 26.0 | 4.7 | 4.2 | 4.1 | 7.7 | 15.4 | 14.7 | 1.1 | 2.7 | 2.6 |
| MXL US | Maxlinear | NR | 1,300 | 14.9 | NA | NA | 1.1 | -0.9 | 0.3 | 13.5 | n.a. | 51.7 | 1.8 | 2.4 | 2.8 | -10.7 | -40.8 | -13.7 | 0.0 | 0.0 | na |
| 3661 TT | Alchip | O | 9,336 | 3,565.0 | 5,000.00 | 42% | 81.5 | 67.3 | 119.2 | 43.7 | 53.0 | 29.9 | 7.2 | 8.2 | 7.3 | 22.4 | 14.6 | 25.8 | 1.2 | 1.0 | 1.7 |
| **Switch System** | | | | | | | | | | | | | | | | | | | | | |
| CLS US | Celestica | NR | 38,452 | 334.6 | NA | NA | 2.4 | 3.9 | 5.9 | 137.7 | 86.2 | 56.5 | 22.5 | 20.5 | 17.3 | 14.2 | 23.3 | 31.7 | 0.0 | 0.0 | 0.0 |
| ANET US | Arista | NR | 169,978 | 135.0 | NA | NA | 1.7 | 2.3 | 2.9 | 77.8 | 59.5 | 47.2 | 23.4 | 17.0 | 13.0 | 34.5 | 33.1 | 30.2 | 0.0 | 0.0 | 0.0 |
| CSCO US | Cisco | NR | 292,378 | 74.0 | NA | NA | 4.1 | 3.6 | 3.9 | 17.9 | 20.7 | 18.8 | 6.7 | 6.4 | 6.1 | 27.1 | 22.6 | 26.3 | 2.1 | 2.2 | 2.2 |
| HPE US | HPE | NR | 30,981 | 23.5 | NA | NA | 2.1 | 2.0 | 1.9 | 11.4 | 11.8 | 12.2 | 1.4 | 1.2 | 1.3 | 10.1 | 11.1 | 11.1 | 2.1 | 2.2 | 2.2 |
| 2345 TT | Accton | O | 18,225 | 1,005.0 | 1,300.00 | 32% | 21.5 | 44.9 | 61.8 | 46.8 | 22.4 | 16.3 | 15.5 | 11.7 | 8.1 | 39.0 | 59.5 | 58.7 | 1.1 | 2.3 | 3.1 |
| **CCL / Substrate / PCB** | | | | | | | | | | | | | | | | | | | | | |
| 2383 TT | Elite Material | N | 15,606 | 1,370.0 | 1,200.00 | -10% | 27.8 | 43.1 | 60.0 | 49.2 | 31.8 | 22.8 | 13.4 | 12.3 | 9.4 | 30.9 | 40.6 | 46.8 | 1.2 | 1.9 | 2.6 |
| 000150 KS | Doosan | NR | 11,208 | 1,021,000.0 | NA | NA | -21,823 | -12,715 | 13,849 | n.a. | n.a. | 73.7 | 8.6 | 9.5 | 12.8 | -17.5 | -11.3 | 15.1 | 0.2 | 0.2 | 0.2 |
| 6274 TT | TUC | NR | 3,549 | 407.0 | NA | NA | 3.1 | 9.6 | 12.2 | 133.4 | 42.6 | 33.4 | 9.5 | 7.8 | 7.5 | 7.0 | 20.1 | 22.8 | 1.0 | 1.6 | 1.9 |
| 6752 JP | Panasonic | O | 27,585 | 1,720.5 | 2,500.00 | 48% | 1.9 | 1.6 | 1.1 | 9.0 | 11.0 | 15.2 | 0.9 | 0.9 | 0.8 | 10.9 | 7.9 | 5.4 | 2.0 | 2.8 | 2.9 |
| 6213 TT | ITEQ | NR | 1,302 | 110.5 | NA | NA | 1.9 | 2.3 | 4.3 | 59.4 | 48.9 | 25.8 | 2.1 | 1.9 | 2.0 | 3.4 | 4.1 | 7.1 | 1.4 | 1.6 | 2.7 |
| 600183 CH | Shengyi | NR | 20,055 | 57.9 | NA | NA | 0.5 | 0.7 | 1.4 | 115.7 | 78.2 | 41.1 | 9.7 | 9.4 | 8.2 | 8.5 | 12.0 | 20.6 | 0.8 | 1.0 | 1.5 |
| 4062 JP | Ibiden | O | 11,796 | 12,820.0 | 11,900.00 | -7% | 2.5 | 2.1 | 2.5 | 50.9 | 60.6 | 51.2 | 3.6 | 3.6 | 3.4 | 7.5 | 6.1 | 6.9 | 0.3 | 0.3 | 0.4 |
| 3037 TT | Unimicron | O | 7,984 | 161.5 | 185.00 | 17% | 3.3 | 3.4 | 8.4 | 48.3 | 47.1 | 19.2 | 2.6 | 2.7 | 2.4 | 5.5 | 5.7 | 13.4 | 0.9 | 0.9 | 2.1 |
| 3189 TT | Kinsus | U | 1,890 | 128.0 | 120.00 | -4% | 0.1 | 3.5 | 6.0 | 1,180.2 | 36.6 | 21.4 | 1.8 | 1.8 | 1.9 | 0.2 | 5.0 | 8.9 | 0.8 | 1.4 | 2.3 |
| 300476 CH | Victory Giant | NR | 35,195 | 288.5 | NA | NA | 0.8 | 1.3 | 5.8 | 369.9 | 215.3 | 49.5 | 32.6 | 27.9 | 17.0 | 9.2 | 13.9 | 36.4 | 0.1 | 0.1 | 0.4 |
| 2368 TT | GCE | NR | 8,969 | 548.0 | NA | NA | 7.3 | 11.5 | 18.9 | 75.6 | 47.5 | 29.1 | 15.8 | 12.5 | 10.1 | 22.7 | 29.4 | 38.7 | 0.6 | 1.1 | 1.5 |
| TTMI US | TTM | NR | 7,285 | 70.5 | NA | NA | 1.3 | 1.7 | 2.4 | 53.0 | 41.2 | 28.9 | 4.8 | 4.6 | 4.5 | -1.2 | 3.7 | /A N/A | 0.0 | 0.0 | na |
| 007660 KS | ISU Petasys | O | 5,273 | 104,000.0 | 98,800.00 | -5% | 11.2 | 21.9 | 38.0 | 93.1 | 47.6 | 27.4 | 21.0 | 10.3 | 7.6 | 24.9 | 28.9 | 31.9 | 0.1 | 0.1 | 0.1 |
| 2313 TT | Compeq | NR | 3,274 | 83.9 | NA | NA | 3.5 | 4.7 | 5.2 | 24.0 | 17.9 | 16.1 | 2.5 | 2.2 | 2.2 | 10.7 | 13.4 | 13.6 | 1.8 | 2.9 | 3.2 |
| 002916 CH | SCC | NR | 19,412 | 207.6 | NA | NA | 2.1 | 2.8 | 4.8 | 98.9 | 73.7 | 43.3 | 10.5 | 9.5 | 7.3 | 11.0 | 13.5 | 19.3 | 0.3 | 0.6 | 0.8 |
| 4958 TT | ZDT | NR | 5,120 | 144.5 | NA | NA | 6.6 | 9.7 | 9.3 | 22.1 | 14.9 | 15.5 | 1.4 | 1.3 | 1.3 | 6.4 | 9.0 | 7.5 | 2.3 | 3.3 | 3.2 |
| **Transceiver/FAU** | | | | | | | | | | | | | | | | | | | | | |
| 300308 CH | Innolight | O | 75,004 | 491.8 | 700.00 | 43% | 4.7 | 10.3 | 24.8 | 104.9 | 47.6 | 19.8 | 26.7 | 16.9 | 9.8 | 25.5 | 35.5 | 49.5 | 0.1 | 0.2 | 0.5 |
| 300502 CH | Eoptolink | NR | 45,086 | 329.1 | NA | NA | 0.7 | 2.9 | 8.9 | 475.0 | 115.2 | 36.8 | 59.8 | 39.2 | 19.5 | 13.4 | 41.1 | 56.9 | 0.0 | 0.1 | 0.3 |
| FN US | Fabrinet | NR | 16,032 | 447.5 | NA | NA | 7.9 | 9.8 | 11.3 | 56.8 | 45.7 | 39.6 | 10.1 | 8.6 | 7.4 | 18.3 | 18.1 | 19.6 | 0.0 | 0.0 | 0.0 |
| COHR US | Coherent | NR | 24,621 | 156.7 | NA | NA | 2 | 3 | 4 | 103.8 | 55.4 | 36.5 | 3.8 | 4.5 | 3.8 | -6.4 | -3.1 | 5.1 | 3.8 | 0.0 | 0.0 |
| LITE US | Lumentum | NR | 17,995 | 253.8 | NA | NA | 2.0 | 1.0 | 4.0 | 126.3 | 267.2 | 64.0 | 14.6 | 16.5 | 15.7 | -27.7 | -22.4 | 20.0 | 0.0 | 0.0 | 0.0 |
| 002281 CH | Accelink | NR | 6,640 | 57.7 | NA | NA | 0.8 | 0.9 | 1.3 | 69.6 | 67.9 | 43.5 | 5.1 | 4.8 | 4.6 | 8.5 | 7.5 | 10.6 | 0.4 | 0.5 | 0.6 |
| AAOI US | AAOI | NR | 1,635 | 23.9 | NA | NA | -0.4 | -0.8 | -0.4 | n.a. | n.a. | n.a. | 4.3 | 5.2 | 3.6 | -28.1 | -84.1 | 6.7 | 0.0 | 0.0 | 0.0 |
| CIEN US | Ciena | NR | 29,554 | 209.5 | NA | NA | 2.7 | 1.8 | 2.6 | 76.7 | 116.0 | 79.4 | 10.6 | 10.6 | 10.4 | 8.1 | 4.6 | 14.1 | 0.0 | 0.0 | 0.0 |
| 300394 CH | TFC | NR | 17,578 | 164.0 | NA | NA | 0.9 | 1.7 | 2.9 | 173.9 | 94.7 | 56.6 | 39.8 | 32.0 | 23.2 | 25.1 | 37.5 | 40.2 | 0.3 | 0.4 | 0.6 |
| 9069 JP | Senko | NR | 2,267 | 1,997.5 | NA | NA | 103.7 | 111.1 | 107.8 | 19.3 | 18.0 | 18.5 | 1.7 | 1.6 | 1.5 | 9.5 | 9.4 | 9.8 | 1.9 | 2.2 | 2.5 |
| 3363 TT | FOCI | NR | 1,220 | 357.5 | NA | NA | 0.1 | -0.5 | 0.8 | 2,750.0 | n.a. | 458.3 | 16.5 | 14.6 | 14.7 | 0.6 | -2.1 | 3.2 | 0.0 | 0.0 | 0.2 |
| 3163 TT | Browave | NR | 601 | 225.5 | NA | NA | 5.6 | 5.8 | NA | 40.1 | 39.2 | na | 6.9 | 6.4 | na | 18.1 | 17.0 | NA | 1.6 | 1.3 | na |
| 5803 JP | Fujikura | NR | 39,736 | 20,075.0 | NA | NA | 107.4 | 269.4 | 498.9 | 187.0 | 74.5 | 40.2 | 17.1 | 14.2 | 11.7 | 16.7 | 22.4 | 28.7 | 0.2 | 0.4 | 0.9 |
| GLW US | Corning | NR | 76,314 | 89.0 | NA | NA | 1.7 | 2.0 | 2.5 | 52.4 | 45.4 | 35.3 | 6.3 | 6.8 | 6.4 | 4.9 | 4.6 | 18.5 | 1.3 | 1.3 | 1.3 |
| **InP substrate/LD/PD/Epi** | | | | | | | | | | | | | | | | | | | | | |
| 3081 TT | Landmark | O | 1,222 | 410.0 | 480.00 | 17% | -0.6 | 4.0 | 10.0 | -688.7 | 103.7 | 41.0 | 10.0 | 9.1 | 7.5 | -1.4 | 8.8 | 18.4 | 0.1 | 0.1 | 0.2 |
| 2455 TT | VPEC | O | 888 | 149.0 | 236.00 | 64% | 3.6 | 2.9 | 5.8 | 41.1 | 50.9 | 25.7 | 8.3 | 8.4 | 7.6 | 20.2 | 16.5 | 29.7 | 2.2 | 2.7 | 2.7 |
| 5802 JP | Sumitomo Elec. | NR | 32,451 | 5,955.0 | NA | NA | 172.2 | 238.8 | 309.0 | 34.6 | 24.9 | 19.3 | 2.2 | 2.0 | 1.9 | 7.0 | 8.3 | 9.4 | 1.1 | 1.6 | 2.0 |
| 4991 TT | GCS | NR | 611 | 163.5 | NA | NA | -7.2 | -2.1 | 0.2 | n.a. | n.a. | 883.8 | 6.1 | 6.1 | 6.1 | -23.7 | -7.9 | -0.3 | 0.0 | 0.0 | na |
| 5801 JP | Furukawa | NR | 4,928 | 9,668.0 | NA | NA | -14.8 | 417.1 | 557.4 | n.a. | 23.2 | 17.3 | 2.1 | 2.0 | 1.9 | 3.1 | 8.0 | 10.6 | 0.7 | 1.1 | 3.2 |
| 688498 CH | Yuanjie | NR | 6,716 | 579.9 | NA | NA | 0.3 | -0.1 | 1.7 | 2,147.7 | n.a. | 332.7 | 23.2 | 23.8 | 22.5 | 0.9 | -0.3 | 6.8 | 0.0 | 0.0 | 0.1 |
| 6503 JP | Mitsubishi Elec | O | 59,314 | 4,297.0 | 5,000.00 | 18% | 1.4 | 1.6 | 1.8 | 31.7 | 27.6 | 23.2 | 2.4 | 2.3 | 2.1 | 8.2 | 8.4 | 9.4 | 1.2 | 1.2 | 1.3 |
| AXTI US | AXT | NR | 506 | 11.0 | NA | NA | -0.3 | -0.2 | -0.4 | n.a. | n.a. | n.a. | n.a. | 2.6 | na | -8.4 | -3.1 | -8.8 | 0.0 | 0.0 | na |
| 5016 JP | JX | NR | 11,244 | 1,940.5 | NA | NA | NA | NA | 91.4 | na | na | 21.2 | 3.0 | 2.9 | 2.7 | NA | 12.8 | 11.9 | 0.4 | 4.2 | 2.1 |
| 4980 JP | Dexerials | NR | 3,261 | 2,363.0 | NA | NA | 108.4 | 162.1 | 151.8 | 21.8 | 14.6 | 15.6 | 5.0 | 4.3 | 4.0 | 29.0 | 30.1 | 25.6 | 1.0 | 2.2 | 2.6 |
| **Equipment** | | | | | | | | | | | | | | | | | | | | | |
| 2360 TT | Chroma | O | 10,184 | 741.0 | 1,000.00 | 36% | 12.5 | 26.6 | 28.1 | 59.3 | 27.9 | 26.4 | 12.6 | 11.1 | 8.8 | 22.5 | 42.2 | 37.2 | 0.9 | 0.9 | 1.4 |

*Source: Bloomberg, Macquarie Research 2025. Priced as at 12 November.*

# Initiating on the transceiver value chain

### Innolight (300308 CH, OP, TP: Rmb700 - PER)

- Innolight is a global leader in transceiver technology.
- We expect a 70% transceiver market CAGR over 2024-27E, outgrowing the switch market. We view transceivers as the major solution for high-speed communications.
- We believe Innolight will sustain its strong leadership due to its sourcing capability amid the developing market shortage and its leading SiPh solution.
- We see its overseas capacity (500k/month 800G transceiver in Thailand) helping it secure overseas market share. We estimate ~50% of high-speed transceiver capacity is outside of China.
- We believe concerns that CPO will rapidly replace transceivers are overdone, as our channel checks suggest the CPO manufacturing process has yet to be finalised and we think transceiver use cases remain very broad with optical scale-up and scale-across opportunities still promising for long-term transceiver demand.

**Valuation**: Our Rmb700 target price is based on 22.8x 2027E EPS, supported by 87% EPS CAGR over 2024-27E.

**Risks**: Slower-than-expected transceiver demand; shortage of key components; earlier-than-expected CPO commercialisation

### LandMark (3081 TT, OP, TP: NT$480 - PER)

- LandMark produces InP epiwafers, an upstream product for laser chips used in transceivers.
- We expect strong transceiver demand driven by AI scale-out networking. We forecast a 44% revenue CAGR for the firm over 2025-27 as its CW laser gains market share amid a likely rise in SiPh transceiver adoption in response to shortages of EML, the current main solution used in AI networking transceivers.
- Furthermore, we view LandMark as well positioned in the CPO market, a longer-term potential solution for AI networking.
- We expect its Datacom business to drive 2026E performance, contributing more than 85% of its revenue, and leading to 153% EPS growth for the year.
- The firm experienced a series of setbacks recently in its CW laser business following a substrate ban in China and manufacturing defects caused by its downstream partners. LandMark is transitioning from 2" to 3"; we expect revenue to pick up when it secures 3" epiwafer qualification.

**Valuation:** Our NT$480 target price is based on 35x 2027E EPS, supported by 86% EPS CAGR over 2026-27E.

**Risks:** Slower-than-expected transceiver demand; excessive capacity at leading players; failed or delayed 3" qualification.

### VPEC (2455 TT, OP, TP: NT$236 - PER)

- VPEC produces GaAs and InP epiwafers for optical chips inside the transceiver, working with GCS (4991 TT) on both PD and LD. It also supplies CW laser through a Japanese laser vendor with great exposure to the AI scale-across market.
- We expect strong transceiver demand driven by AI scale-out networking.
- We forecast a 91% revenue CAGR for VPEC's data-centre photonics business over 2026/27E, and see gross margin rising 11ppts over two years to hit ~48% by 2027E.
- We anticipate tighter supply of networking components ahead, as data centre demand ramps with scale-out spec upgrades. We believe second-tier vendors will benefit the most from the spill-over effect. We believe VPEC will obtain more orders in addition to its current track record of shipping through its Japanese customers.

- For its Microelectronics business, we see the Skyworks/Qorvo merger impact as limited, as both companies are already VPEC customers, and the two companies will continue to operate independently in 2026. We believe VPEC will sustain its margin as its major competitor IQE fades with a plan to sale its Taiwan business.

**Valuation:** Our NT$236 target price is based on 30x 2027E EPS, with 64% EPS CAGR over the next two years.

**Risks.** Slower-than-expected transceiver demand; excessive capacity at leading players; declining wireless communication market.

Core idea: To connect as many GPUs as possible with fastest interconnect in a power-efficient way.

# AI parallel computing

Parallel computing has always been a major focus in technology sector. From a hardware perspective, we have come a long way, from single core CPUs to multi-core CPUs, to GPUs with thousands of cores, to processors with multi-die packages, to rack-scale interconnected servers, to GPU server clusters. Every step of parallel computing requires an optimised interconnect. AI computing missions are now far too large for a single chip to handle: it would take a single GPU some 355 years to train GPT3 - not even the latest multi-modal massive model. To serve spiking computing demand, parallel computing is necessary. Following huge investment in silicon upgrades including GPU advancements and advanced packaging, system upgrades and cluster upgrades are emerging as a cost-effective alternative for AI hardware upgrades. AI networking, optimised interconnect technologies in AI cluster systems, inspires huge hardware market opportunity.

- **Booming AI interconnects bigger clusters.** Hyperscalers are evolving their AI capabilities by expanding the networks, incorporating bigger clusters. This is mainly because a model with over a trillion parameters does not fit on a single accelerator; a brand-new network setup is crucial to support that kind of scale. Hyperscale LLM model training deploys a huge number of GPUs beyond a single node (eg, Meta detailed 24k GPU training clusters for Lllama3), then these GPUs are interconnected using high-speed interconnects. Even if the apparent inference traffic per query is smaller than during training, the aggregate traffic at hyperscale inference work, running thousands of concurrent sessions, demands a larger scale of network links to connect more GPUs and inference clusters. Given the physical limitations of AI accelerators, in contrast to surging AI workloads, we believe network expansion will start to take a higher priority in hyperscalers' AI ramp-up plans.
- **Additional networking architecture segments for better coordination and communication.** It is important to note that when clusters scales, different segments of interconnects are deployed to facilitate bigger cluster communication. Scale Up is emerging as an alternative for silicon upgrades to connect GPU within the rack, Scale Out is an optimised connection across racks to work coherently on the same AI training mission, and Scale Across, or so-called data centre interconnect (DCI), is emerging as a new innovative solution for hyperscalers to span beyond intra-data centre networks such as scale-up and scale-out, connecting multiple data centres to create a much bigger AI hardware cluster. This not only helps global hyperscalers reach much farther data centres, but also reduces latency as well as flexible integration of multi-vendor fabrics (ie, GPU clusters, TPU clusters, and AI inference clusters) via better-aligned topology.
- **Multifold bandwidth demand ahead.** Bigger cluster sizes will require additional layers of switches and interconnects, and in an ideal AI scale-out networking setup, each layer of interconnections will have the same bandwidth, ie, similar hardware content value. For instance, three-tier (three-layers) networks require >3x number of switches and 67% more optics compared to two-tier networks, per Broadcom. This should translate into multi-fold TAM growth for network hardware, including network switches and transceivers.

System upgrades and cluster upgrades are emerging as a cost-effective alternative to hardware upgrades . . .

. . . and larger clusters can reduce training time significantly . . .

. . . so better networking will pay for itself

**Q: Why do we need AI networking?**

1. **Silicon upgrades alone are insufficient to service the surge in demand for AI.**

LLMs, photo-generating models, multimodal models, physical AI, all kinds of applications are emerging, and more training is needed (Figure 8). Silicon upgrades alone are not going to maximise computing power. The upgrade in silicon performance is usually marginal, but demand growth is explosive.

**2. 'Divide and conquer' to achieve quicker product launch through clustering.**

An AI GPU cluster requires multiple GPUs to work on the same LLM training mission, and due to the sheer size of the LLM, efforts to break up processes into multiple parallel tasks rather than attacking a monolithic problem head-on ('divide and conquer') is more feasible. Larger clusters can reduce training time significantly. The lead time for each generation of model will determine the winners, and therefore AI companies are racing to build larger and larger clusters.

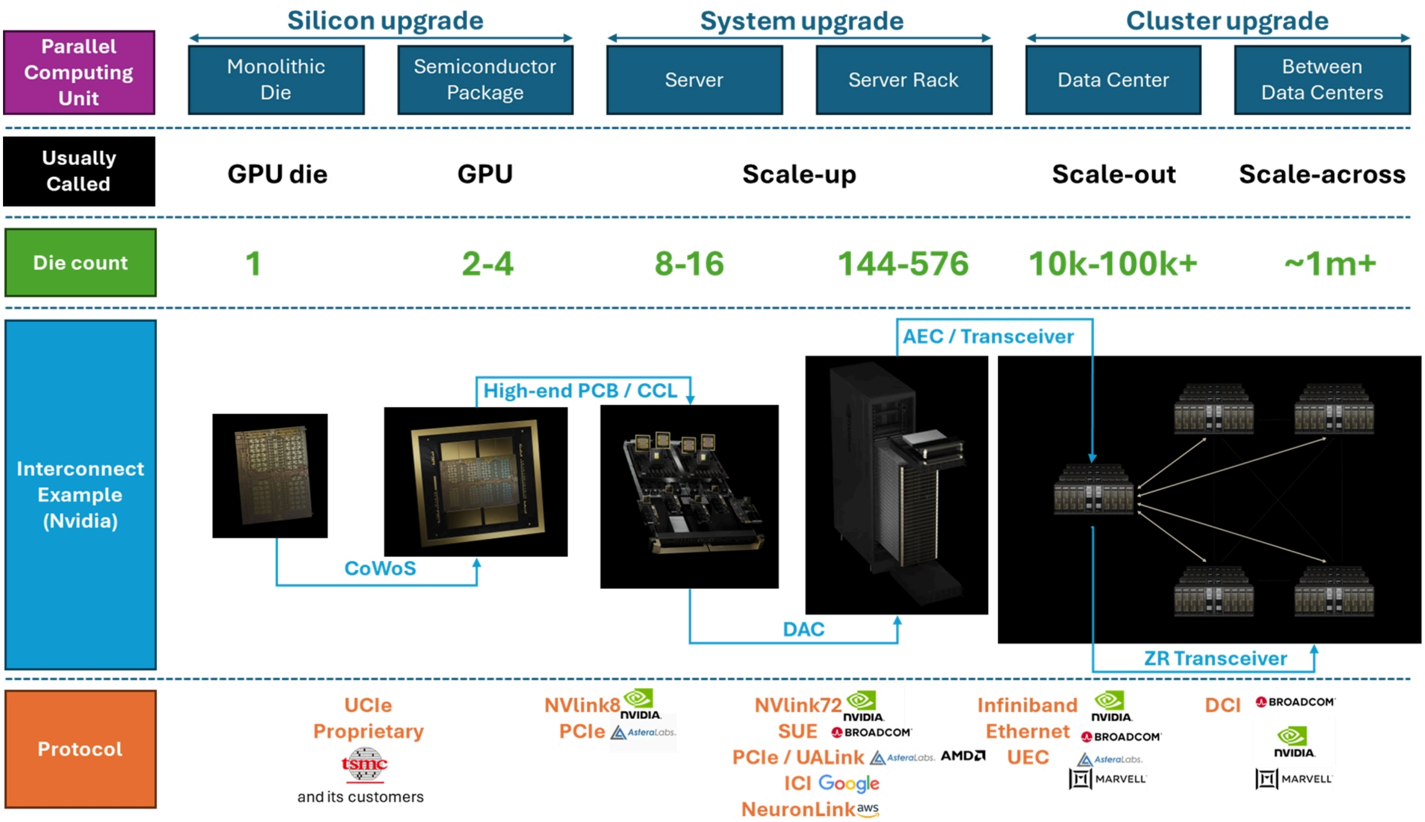
**3. Better networking setup can result in better utilisation for expensive GPUs.**

An optimised division of labour can improve GPU utilisation. According to Meta's Llama 3 white paper, its 24k cluster is running at a poor ~40% GPU utilisation rate due to networking congestion. By dividing the model smartly and allocating the jobs with high-speed networking, it could achieve better training efficiency and better GPU utilisation, which is a core competence in the AI era, ie**,** better networking will pay for itself.

**Q: What are networking protocols and why are they important?**

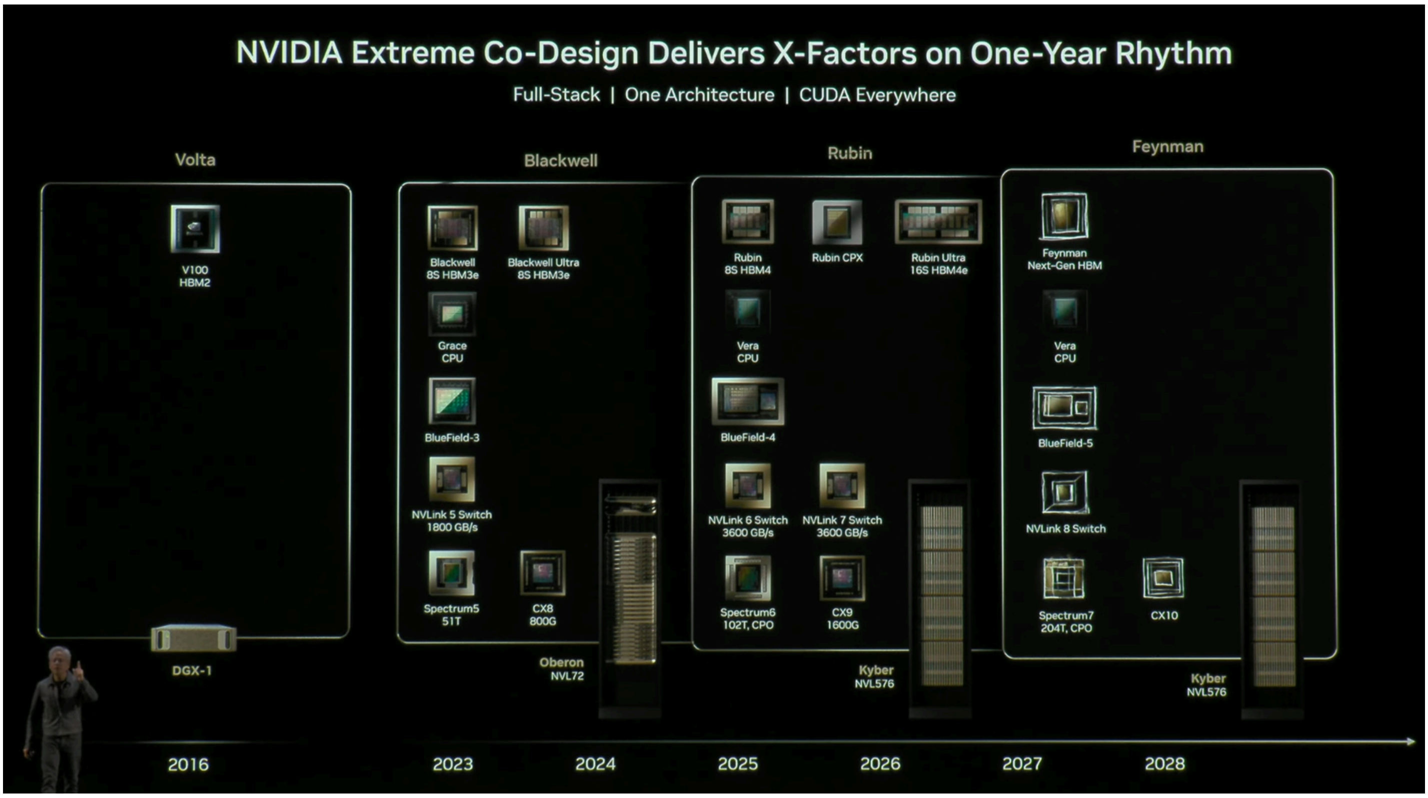
Protocols define which camp/which companies are in the ecosystem. XPUs will need to communicate with each other, and protocols are their language. Switches serve as the logistics centres, and protocols define how messages are sent. For each protocol, there will be a corresponding product from each company; Figure 5 provides an overview, which we will explain in detail in following paragraphs.

**Figure 5 - Interconnect technology overview**



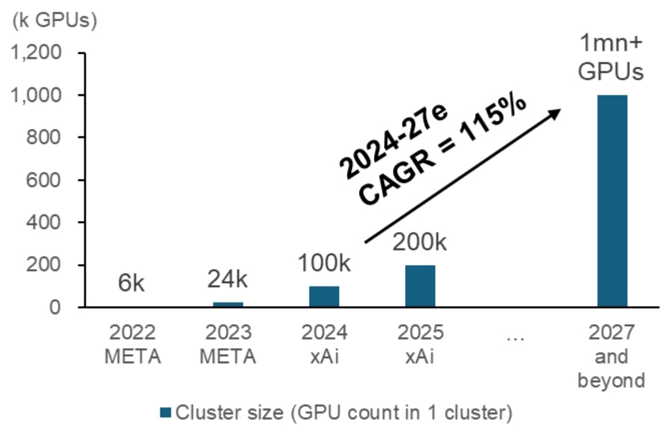

*Source: Companies, Macquarie Research, Nov 2025*

**Figure 6 - Nvidia's full stack solution for parallel computing**



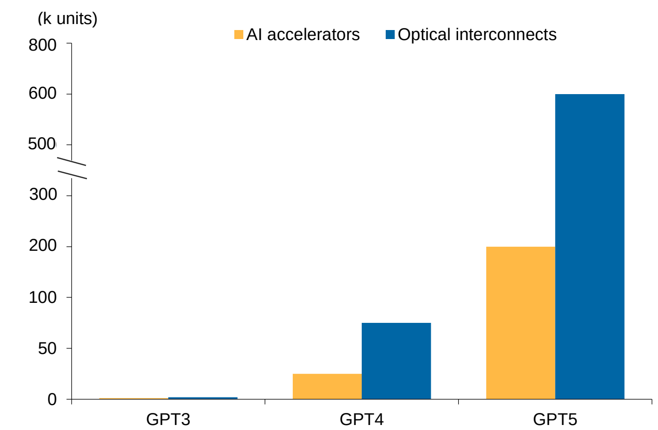

*Source: Nvidia, Nov 2025*

**Figure 7 - GPU cluster size**




*Source: Companies, Macquarie Research, Nov 2025*

**Figure 8 - LLMs requiring bigger clusters with more GPUs and more interconnect**




*Source: Company, Macquarie Research, Nov 2025*

# Concept: Networking typology

**Q: What is the difference vs traditional networking? Where are the opportunities?**

AI networking is hugely different from traditional data-centre networking architecture. (Figure 9).

**Larger clusters enabled by additional segments of Scale Up and Scale Out:** The additional links of Scale Up (GPU fabrics) and Scale Out (Infiniband or Ethernet with RoCE) is necessary for divided computing tasks to sync together. We need better networking architecture/ silicon to handle the traffic. The network's additional segments are the main reason why we think of this networking upgrade as non-cyclical and revolutionary for the industry.

High-speed connection uniformity creates demand for high-speed switches and transceivers

**Uniform high-speed interconnects across all layers**: The higher data-rate is consistent and uniform throughout different layers of the network, which is different from the traditional network**.** The uniformity of high-speed connection creates a huge opportunity for high-speed interconnection, including high-speed switches and transceivers, from copper to optical.

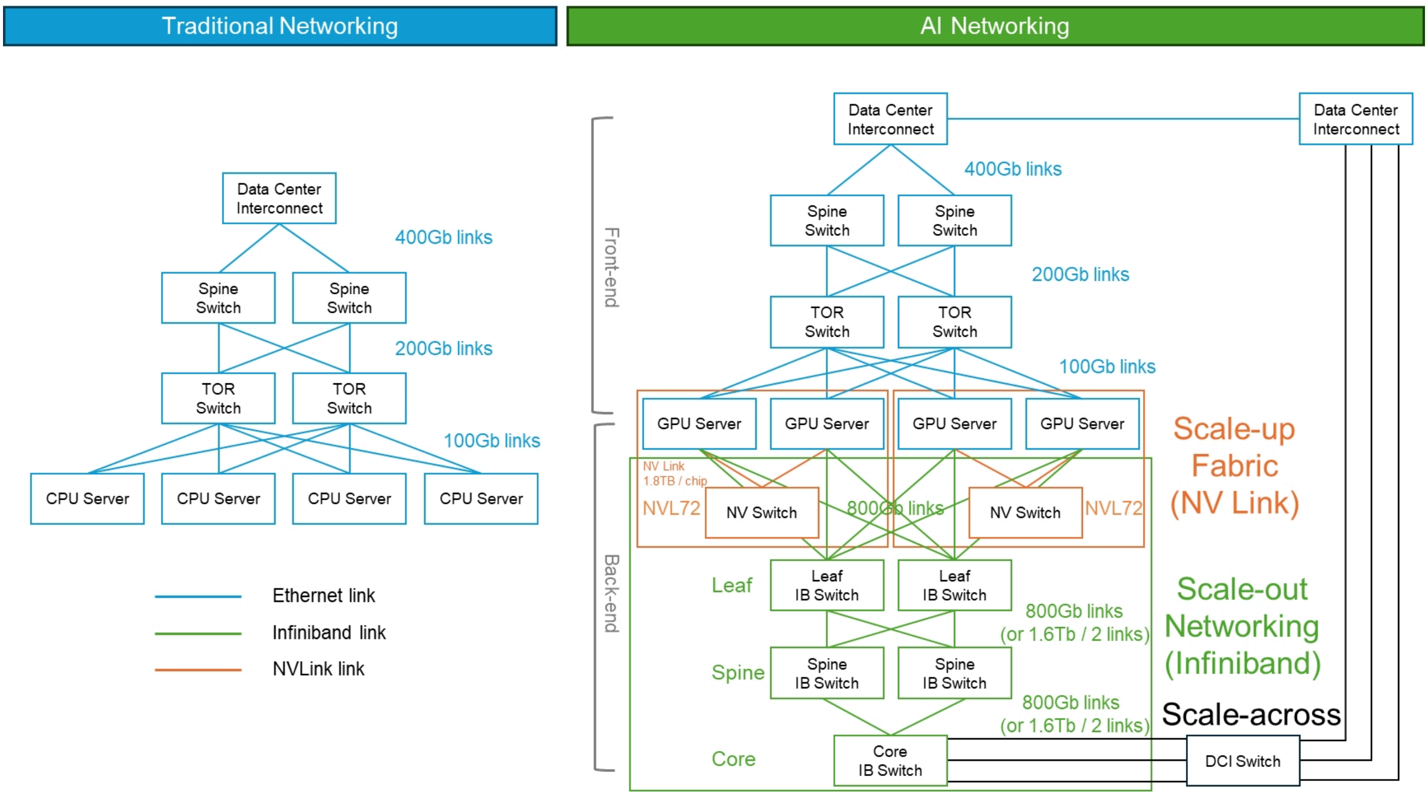
Technologies such as InfiniBand or Ethernet with RoCE are needed and will spur demand for a new generation of NICs and switches.

**GPU utilisation is a key driver**: Standard Ethernet traffic accepts a degree of the packet (a set of dataflow) loss, that means, when dataflow is sent but not delivered, it allows retransmission of lost packets. In AI training, workloads are synchronised across all GPU clusters, performing collective operations. A single dropped packet could delay the completion of an entire training step, leaving vast amounts of GPU compute power idle while waiting for retransmissions. To prevent this, AI networkings are engineered to be lossless fabrics. Technologies such as InfiniBand or Ethernet with RoCE are needed and inspire demand for a new generation of NICs and switches.

We expect creative new interconnect solutions, including LPO, CPO and OCS.

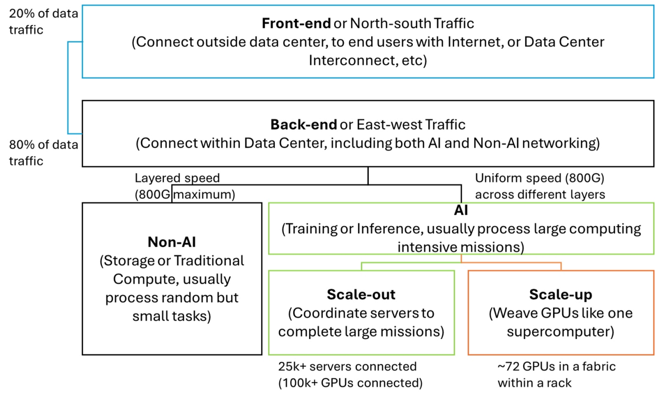
**Power budget still a major constraint:** Increasing traffic also poses a threat to power-hungry AI data centres, with power budget constraints, necessitating more efficient solutions for interconnections. We expect creative new ways of data interconnection to surge, including LPO, CPO and OCS.

**Figure 9 - AI networking architecture overview (Nvidia as example)**



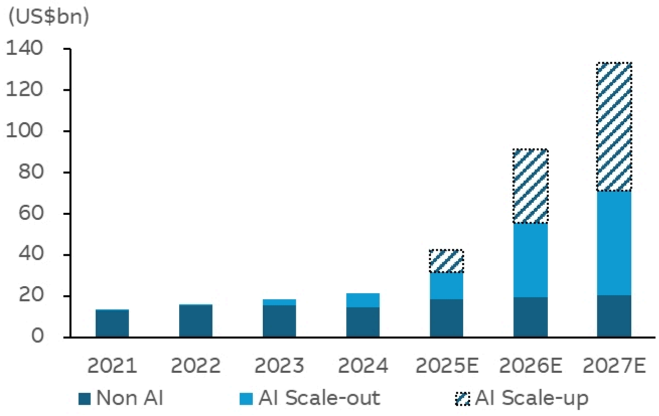

*Source: Nvidia, Macquarie Research, Nov 2025*

**Figure 10 - Typology of networking segments**




*Source: Macquarie Research, Nov 2025*

**Figure 11 - MQ projected different network switch growth**




*Source: Macquarie Research, Nov 2025*

# Estimating AI networking market size

## Bigger AI clusters spark growth

We expect the switch market to deliver a 50% revenue CAGR over 2024-27E, driven by massive AI cluster deployment and scale-networking demand. Transceivers are the solution to high-speed interconnect, and here we forecast a 70% revenue CAGR over 2024-27E. The scale-up market is a proprietary market with different ecosystems built by different vendors, and is difficult for others to penetrate. We are bullish on the long-term demand for networking, as we believe there are opportunities ahead, such as CPO, yet we take a conservative view as the first wave of beneficiaries will be testing-equipment names before any real adoption.

**Figure 12 - Networking market opportunity overview**

| Networking Considerations | Target | Implication | Hardware Market Opportunity |
|---|---|---|---|
| Supercluster | Connect as many GPUs as possible to work on a huge mission. | Additional segments of scale-up, scale-out, scale-across are needed for AI cluster. | Major CSP's cluster size to grow 115% in 2024-27e, reaching 1mn+ in 2027, inspiring the networking market to grow multi-fold. |
| Faster Interconnect | High-speed across all layers of networking , not just in core but also on leaf and spine. | Massive amount of high-speed 800G, 1.6T transceivers / AEC are needed | Transceiver market to grow at 70% in 2024~2027e |
| GPU utilization | Better switching. Advanced traffic control to achieve higher GPU utilization | Next-gen switch and smart-NIC are upgraded alongside with GPU cadence. | We expect We believe the scale-out switch market to grow 50% in next 3 years, with Nvidia leading the transiiton |
| Power Efficiency | Save more power. Less active component or conversion | SiPh, LPO, CPO, OCS are introduced to decrease power consumption. | We believe the first wave beneficiaries of CPO will be testing equipment makers before mass production ramps. |

*Source: Macquarie Research, Nov 2025*

The additional segments needed in AI networks are a multiplier for networking components demand

**Non-cyclical growth ahead**

We are seeing networking bandwidth continue to grow. (Figure 13). But the traditional networking industry has been very cyclical due to price erosion from switch-speed migration. (Figure 14). Not anymore for AI. As Figure 9 illustrates, AI networking introduces additional segments: the scale-out network, which runs at top data rate across all layers, and scale-up fabric, which is even denser and faster than the scale-out network. These two segments result in exponential demand for networking components as AI demand grows, and we anticipate a tectonic shift across the entire supply chain as it responds.

The irregular demand patterns have already shocked the supply chain, as we are seeing explosive demand for all AI networking components, with shortages of PCB, CCL, and transceiver key components such as EML laser. We expect this to continue as we move into 2026; we are still projecting at least another three waves of bandwidth demand, driven by CX8 800G upgrade, ASIC ramps, and further clustering. In the future, we are positive on 1)scale-across demand driven by super clusters; 2)optical scale-up, and we view CPO as a possible solution, yet it is still early to estimate the adoption rate and identify winners prior to mass production.

**Our scale-out AI networking assumptions**

For scale-out, we estimate a 100%+ AI cluster size from 2024-27E CAGR, mainly driven by the deployment of AI GPUs and ASICs in superclusters. Further evidence of the speed with which scale-out is expanding can be found in Nvidia's abnormal Networking Interface Card (NIC) upgrade speed - after having stayed at 400G for some three years (in line with the industry cadence), it now aims to upgrade from 800G to 1.6T within only a year.

In calculating the market size of AI scale-out networking switch systems, we begin by estimating the total bandwidth requirements, which is essentially the sum of switch ports needed across GPU deployments. The first key assumption is ports per GPU, which is a reflection of cluster size, as larger clusters requires more layers of switches. For port speed, we obtain the spec of the corresponding NIC speed for each generation of GPU, which will serve as the underlying speed of the entire AI networking across all layers. Finally, we account for port speed migration, as a portion of switch ports used in higher network layers (spine or core) will usually be combined into higher speed ports, which keep the bandwidth constant but reduce the ports and switch count.

The scale-up market is multiples larger than the scale-out market

We forecast a 50% switch market CAGR over 2024-27E

**For scale-up fabric**

The switch density per GPU and the bandwidth is far higher in scale-up racks than in scale-out clusters, in the order of ~3x or even 6x+. In scale-out network, we think the density is roughly 2-3 switches per 32 GPUs, but in NVL72 scale-up fabric, it is 18 switches per 72 GPUs, which is much higher. We might be able to size the scale-up market as a multiple of scale-out market, as it is on a magnitude of several times the size of the scale-out market. While it is clear that it is a much larger market than scale-out, it is much harder to forecast as currently only Nvidia's proprietary NVLink is being deployed, and it is always viewed as part of the server rack, not networking, due to its full integration and proprietary nature. To date, there has been only limited deployment of scale-up systems in data centres. Broadcom is planning to enter the market with a scale-up switch, yet we haven't seen wider adoption in 2025. UALink is even further away than Broadcom, with a switch chip launch date in 2H26. We can only posit a penetration rate for a possible set-up rather than a precise figure**.**

We believe It is better to separate the market size by each Switch provider due to proprietary protocols. Our current scale-up market forecast is based on estimates for Nvidia, Broadcom, and UALink, based on the switch:XPU ratio as the underlying assumption. We start from Nvidia's market size as a gauge, on which basis we estimate the possible market size for Broadcom will be around ~US$14.5bn, and we assume the first deployment of broadcom scale-up system should be in 2026. Another player is Astera Labs, whose major customers are AWS and AMD; we assume its product will be ready in 2H26 and mass production in 2027, with a market size of US$8.2bn in 2027.

We estimate a 70% transceiver market CAGR over 2024-27E

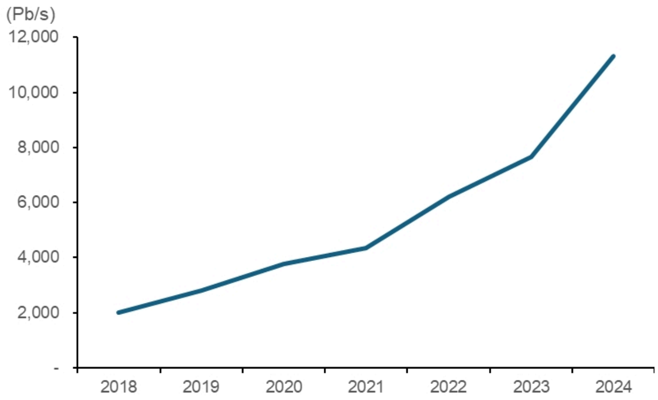
**Transceiver demand**

We have modeled in the transceiver attach rate / per GPU, and we forecast a higher attach rate due to larger cluster sizes across all major CSPs. We also considered demand from traditional networking (front-end and non-AI back-end), scale-out, and some use cases from scale-up, as well as taking into consideration potential adoption of AEC in short-distance interconnect and CPO in high-speed interconnect for scale-out in 2025-27E.

**CPO**

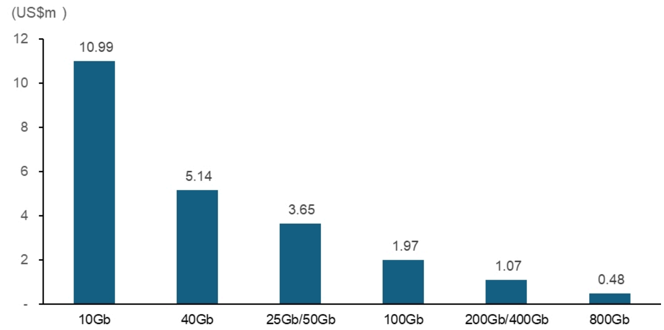
We are convinced that CPO will be a mainstream solution going forward, but we are conservative in our assumptions of the timing of CPO adoption, as our channel checks suggest that the manufacturing process for CPO is yet to be finalised. We believe the first-wave beneficiaries will be testing equipment makers, as it is essential for semiconductor industry to determine the equipment and manufacturing processes before going into mass production. We assumed some CPO penetration in 1.6T ports and potential erosion for the transceiver market in 2027E, and we believe the real adoption and production ramp will start in 2028E at the earliest.

**Figure 13 - IDC historical networking switch bandwidth**



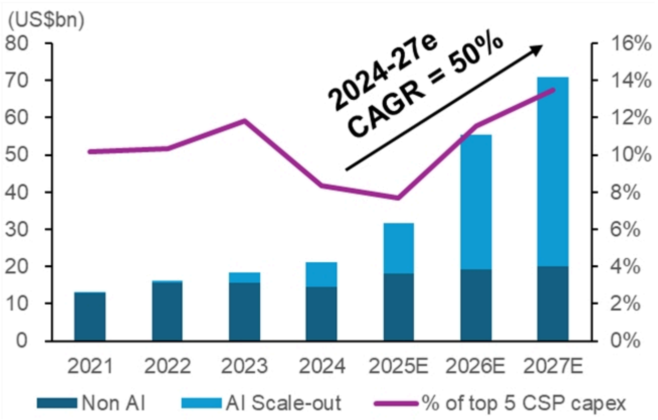

*Source: IDC, Macquarie Research, Nov 2025*

**Figure 14 - Switch bandwidth cost per Pb/s**



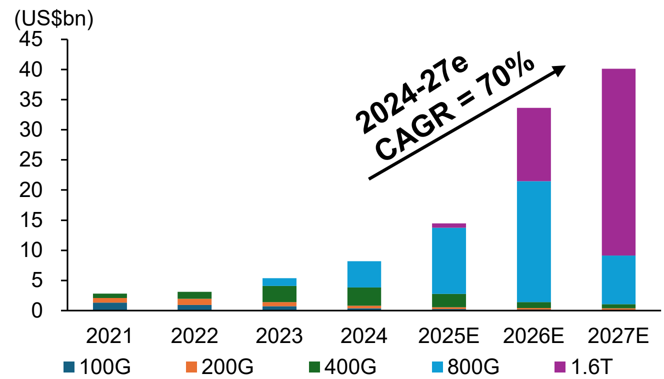

*Source: IDC, Macquarie Research, Nov 2025*

**Figure 15 - AI scale-out switch market growth**



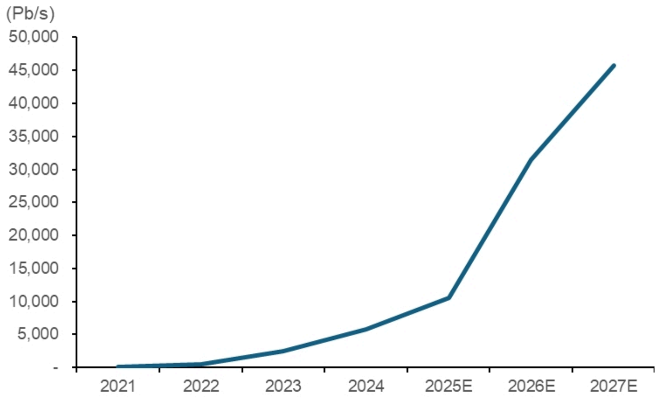

*Source: IDC, Macquarie Research, Nov 2025*

**Figure 16 - MQ transceiver market forecast**



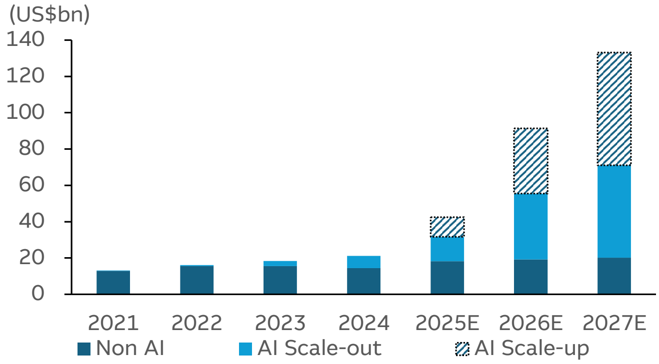

*Source: IDC, Macquarie Research, Nov 2025*

**Figure 17 - MQ AI scale-out bandwidth demand**




*Source: IDC, Macquarie Research, Nov 2025*

**Figure 18 - MQ networking switch market forecast - include scale-up**




*Source: IDC, Macquarie Research, Nov 2025*

# Scale-out networking overview

Scale-out networking is the underlying setup for clusters. We have seen acceleration in Nvidia's roadmap of their CX-series smart NIC cards, from CX7 400G to CX9 1.6T within 3 year. We view this as a clear evidence of the scale-out growth trajectory. We expect three waves of revenue growth for AI networking hardware, with the first being the GB300 ramp-up in 4Q25, a second being the ASIC upgrade to 800G in 1H26E, and a third the Rubin ramp-up and even larger ASIC buildouts in 2H26E.

**Clustering and cluster size is limited by switch capacity.** For a 64-port network switch, only 32 GPUs will be connected to that switch, as the remaining ports will need to connect to upper layer switches. But switch capacity is limited by semiconductor nodes and SerDes (serialiser/deserialiser) design, which are both very difficult to progress. Another way to go is to add another layer of switches and interconnect as clusters get bigger, but this will result in additional spending on networking per GPU, as well as leading to higher latency.

The smartNIC bandwidth serves as the underlying speed across the full AI scale-out network, a difference from the traditional networking setup. If smartNIC data rates double, the whole AI networking scale-out bandwidth will double. Nvidia's accelerating pace on upgrading its Connect-X series from CX7 (400G) to CX9 (1.6T) in just 3 years, will inevitably lead to a spike in demand for AI networking.

Another important thing for clustering is network control. The algorithm, firmware and digestion control, such as RoCE, need to work hand-in-hand, creating barriers other than just hardware performance. Proprietary protocols such as Infiniband offer better performance, but are bound by ecosystem.

We highlighted some mainstream solutions:

**NVIDIA: The Full-Stack, Dual-Pronged Approach**

NVIDIA offers two distinct, end-to-end solutions for AI scale-out, leveraging its full control over the GPU, NIC, switch, and software stacks

- **1. The InfiniBand Solution (Quantum X):**
  - ⇒ **Switch:** The NVIDIA Quantum InfiniBand switch platform is the mature, widely deployed solution. It is a lossless-by-design protocol, making it well-suited for the most demanding AI workloads.
  - ⇒ **Port Speed:** Quantum-2 offers 800Gbps per port (NDR). The upcoming Quantum-3 platform is designed for 1.6Tbps (XDR).
  - ⇒ **NIC:** The ConnectX-8 SmartNIC is the standard for 800G InfiniBand. For next-generation clusters, the ConnectX-9 is being introduced to support 1.6T speeds.
- **2. The Ethernet Solution (Spectrum-X):**
  - ⇒ **Switch:** The NVIDIA Spectrum-4 is the flagship Ethernet switch, purpose-built for AI. It combines the switch with the BlueField-3 DPU to optimise traffic flow and congestion control for RoCE, creating what NVIDIA markets as a "Super-Ethernet."

- ⇒ **Port Speed:** Spectrum-4 delivers 800Gbps per port. The upcoming Spectrum-6 platform will enable 1.6Tbps ports.
- ⇒ **NIC:** The ConnectX-8 SmartNIC (800Gpbs) and BlueField-3 DPU are the core components on the server side, tightly integrated with the Spectrum switch to provide adaptive routing and prevent the "long-tail latency effect."

**Broadcom: The Merchant Silicon Powering the Ecosystem**

Broadcom does not sell complete systems but provides the foundational switch chip that powers the majority of high-performance Ethernet switches from vendors like Arista, Dell, Cisco, and Juniper, as well as white-box ODMs such as Celestica and Accton.

- **Switch (Silicon):**
  - ⇒ **Tomahawk 5:** This is the current, widely deployed industry workhorse. It is a 51.2T switch chip that enables vendors to build high-density switches with 64 ports of 800Gbps.
  - ⇒ **Tomahawk 6:** As of late 2025, the next-generation Tomahawk 6 (102.4T) is the cutting edge. Systems built on this chip are beginning to ship from major vendors, enabling the first wave of **1.6Tbps** port deployments.
- **Port Speed:** Tomahawk 5 underpins the 800Gbps ecosystem, while Tomahawk 6 is driving the transition to 1.6Tbps.
- **NIC:** Broadcom launched its high-performance AI NIC Thor Ultra (800Gbps). Its switch silicon is designed to work with NICs, but as the AI workloads become more complex than ever, Broadcom has chosen to enter the smart NIC field as well.

**Amazon (AWS): The Custom Hyperscale Fabric**

Amazon builds its own networking hardware and software, optimised for AWS services and its custom AI accelerators (Trainium and Inferentia).

- **Switch:** AWS uses custom-designed switches, often powered by its own ASICs or semi-custom chips. These are not commercially available but are deployed by the thousands within its data centres to create petabit-scale fabrics.
- **Port Speed:** While exact internal speeds are not always public, AWS' high-performance clusters utilise a fabric with link speeds equivalent to or exceeding the 800Gbps commercial standard.
- **NIC:** The core of its solution is the Elastic Fabric Adapter (EFA), a custom NIC and associated software stack. EFA provides a reliable transport protocol on top of Ethernet, allowing AI applications to bypass the operating system's kernel for low-latency, high-bandwidth direct communication—achieving a similar outcome to RoCE or InfiniBand.

**Google (GCP): The leader in Optical networking technology**

Google has been co-designing its hardware and software for over a decade, with its networking built specifically to serve its Tensor Processing Units (TPUs).

- **Switch:** Google employs a unique, hybrid approach using both traditional packet switches and its ow**n Optical Circuit Switches (OCS).** This allows it to physically reconfigure the network topology on the fly using light paths, creating direct, ultra-low-latency connections between racks of TPUs that are optimised for a specific AI model's communication patterns.
- **Port Speed:** the major advantage of OCS is that the port speed is only related to transceiver speed, and any upgrade in transceiver will not require an upgrade in the switch, which makes the speed migration cost much easier.

## Figure 19 - ASIC (Network switch IC) Usage in the tier 1 cloud

| ASIC Size (Tbps) | Process Node | SERDES Technology | Leaf Port Speed | Spine/ Core Port Speed | Product Cycle |
|---|---|---|---|---|---|
| 1.3 Tbps | 65/40nm | 10 Gbps | 10 Gbps | 10/40 Gbps | 2012-2016 |
| 3.2 Tbps | 28nm | 25 Gbps | 25/50 Gbps | 100 Gbps | 2017-2021 |
| 6.4 Tbps | 16nm | 25 Gbps | 25/50 Gbps | 100 Gbps | 2018-2021 |
| 12.8 Tbps | 16nm | 50 Gbps | 50/100 Gbps | 200/400 Gbps | 2018-2026E |
| 25.6 Tbps | 7nm | 100 Gbps | 100/200 Gbps | 200/400 Gbps | 2022-2025E |
| **51.2 Tbps** | **5nm** | **100 Gbps** | **400/800 Gbps** | **400/800 Gbps** | **2024-2027E** |
| **102.4 Tbps** | **3nm** | **200 Gbps** | **800G/ 1.6T** | **1.6 Tbps** | **2025-2030E** |
| 204.8 Tbps | 2nm | 400 Gbps | 1.6T/ 3.2T | 3.2 Tbps | 2026- |

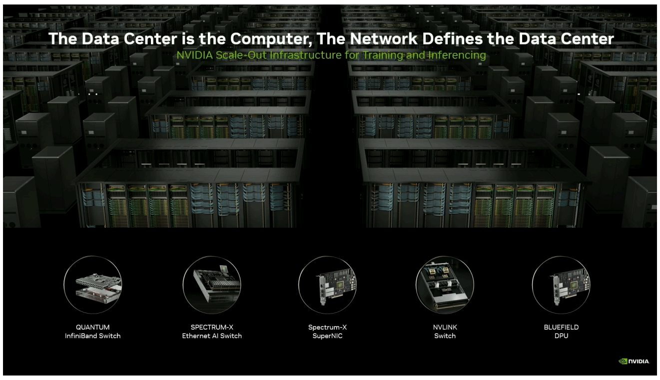
*Source: 650 Group, Nov 2025*

## Figure 20 - Network switch IC launch timeline by main brands

| | Form Factor | Process Node | Announcement | Original Shipment Date | Actual Shipment Date |
|---|---|---|---|---|---|
| **800Gbps network switch IC** | | | | | |
| Nvidia Spectrum 4 | 512 x 112G | 4nm (Nvidia Line) | March. 2022 | Late 2022/ Early 2023 | Late Summer 2024 |
| Broadcom Tomahawk 5-100 | 512 x 112G | 5nm | August. 2022 | Fall 2022 | 2H24 |
| Cisco | 512 x 112G | 5nm | June. 2023 | Fall 2023 | |
| Innovium (Marvell) Teralynx 10 | 512 x 112G | 5nm | March. 2023 | Summer 2023 | Summer 2024 |
| **1.6Tbps network switch IC** | | | | | |
| Nvidia Spectrum 6 | 512 x 224G | 3nm | TBD | | |
| Broadcom Tomahawk 6-100 | 1024 x 112G | 3nm | June. 2025 | | |

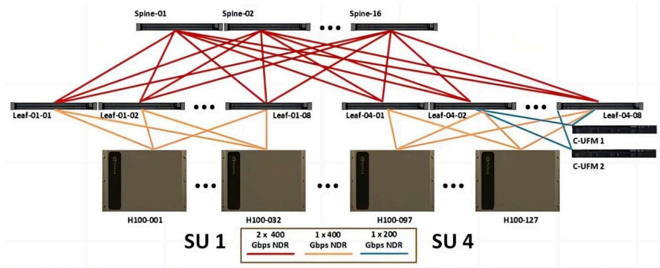
*Source: 650 Group, Macquarie Research, Nov 2025*

## Figure 21 - Nvidia networking product portfolio



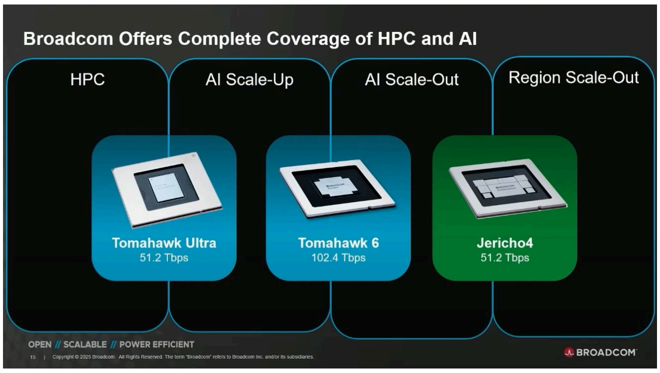

*Source: Nvidia, Nov 2025*

## Figure 22 - H100 scale-out setup



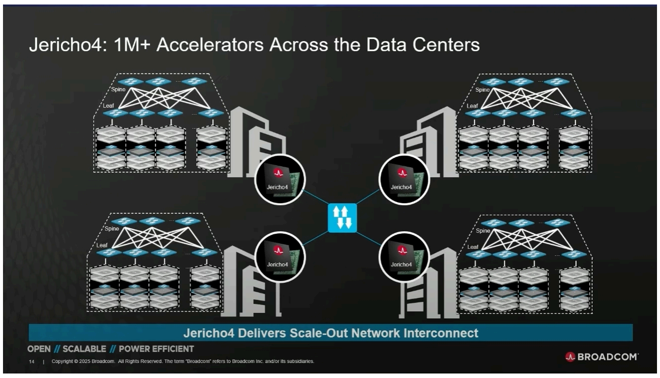

*Source: Nvidia, Nov 2025*

## Figure 23 - Broadcom networking product line-up



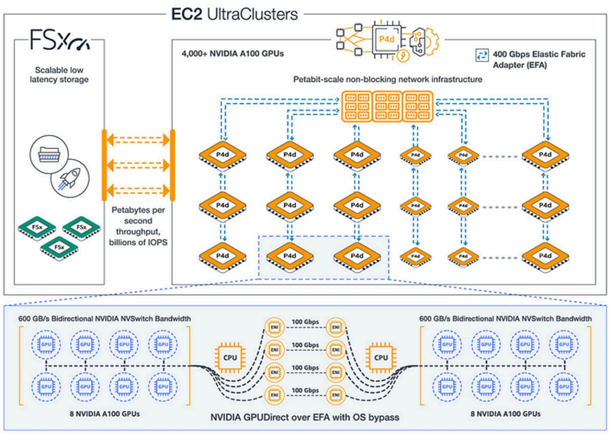

*Source: Broadcom, Nov 2025*

## Figure 24 - Broadcom scale-out switch setup



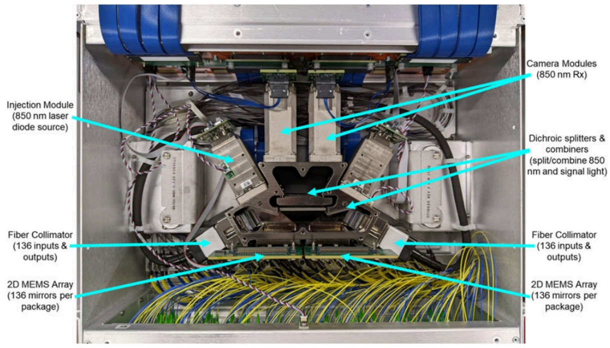

*Source: Broadcom, Nov 2025*

## Figure 25 - Amazon scale-out network overview




*Source: Amazon, Nov 2025*

## Figure 26 - Google's Optical Circuit Switch




*Source: Google, Nov 2025*

# Scale-up fabric overview

Scale-up fabrics are more dense than scale-out networks, requiring more hardware...

When we compare scale-up with scale-out, we note that scale-up networking bandwidth is several times higher than scale-out, due to its interconnection density. The links per GPU for scale-up is 18 links, compared to just 3 links in scale-out. That's why, when the industry talks about scale-up, it describes it in terms of a "fabric", as opposed to a "network" for scale-out, fabrics being much more densely woven than nets.

...but with little tolerance for latency or translation it is dominated by proprietary protocols

The challenge for scale-up is actually not on the switch side, but on the XPU side. How to insert enough SerDes I/O into XPUs, without exponentially increasing silicon cost, performance, and power consumption is the key.

The SerDes on XPU side needs to work seamlessly in the same protocol on the switch side. NVLink is an example. No other company can enter Nvidia's scale-up domain, purely because NVLink is a proprietary protocol. Scale-up fabric needs ultra-low latency and does not allow room for protocol translation.

Basically the scale-up system is:

**1) an integrated supercomputer** (Figure 28)

**2) with shared memory access** (Figure 30)

**3) within a rack, but not always** (Figure 29)

In the past, computer upgrades were mainly silicon upgrades. Now as it becomes ever more challenging to keep up with Moore's law, system upgrades have started to prevail.

Conceptually, the difference between scale-up and scale-out is that one is in-rack and one is between-rack connections. But the truly defining feature of scale-up is the shared memory setup, enabled by ultra-fast, ultra low latency interconnect fabrics. If we observe the spec sheet of NVlink, it's notable that specs are defined in Gigabytes, the usual units for memory, rather than Gigabits, the usual unit for networking. After all, the scale-up setup is like a supercomputer.

**Proprietary protocol: XPU SerDes need to be compatible with Switch SerDes.**

Scale-up fabric is usually proprietary, due to its latency requirement. Whereas communication between scale-out networking allows for translation and latency, interconnect between scale-up fabrics are sensitive to latency, making no room for protocol interpretation, and therefore, proprietary. **The SerDes protocol in XPU dies needs to be exactly the same protocol as used in the switch die.**

Direct XPU links mean that the GXPU package should include enough SerDes I/O, which will take up die area, driving up cost and TDP. SerDes quality relates to die size, power consumption, signal quality, among other factors. On the switch side, scale-up is relatively easy; the challenges lie on the XPU side.

The winner must have not only best SerDes, but also will need the ability to integrate SerDes IP into the XPU, which we believe, only Nvidia and Broadcom are capable of. For UALink, without a 224Gbps SerDes provider in the alliance, it will struggle to design a SerDes, let alone integrate SerDes IP on the XPU side, given the already limited die area.

As power consumption is already a key issue for data centres, most existing scale-up is using passive interconnect, which is the main reason why Nvidia is inserting as many XPUs as possible within the same rack.

**Nvidia: The dominant incumbent**

Nvidia's solution is the market benchmark, evolving from technology originally developed for HPC supercomputers long before the generative AI boom.

- **Approach:** A completely vertically integrated and proprietary stack, ensuring maximum performance through co-design of the GPU, switch, and software.
- **Switch Product:** The **NVSwitch 5rd Generation** is the core of the Blackwell platform. These are dedicated, stand-alone chips that connect the GPUs.
- **Protocol: NVLink 5th Generation**. This protocol provides direct, high-speed peer-to-peer communication between GPUs.
- **GPU Fabric Size:** A single NVL72 rack creates a 72-GPU NVLink domain. The architecture can connect up to eight of these racks, enabling a massive **576-GPU** domain where every GPU can communicate with every other GPU at full speed.
- **Processor SerDes:** The NVLink 5 SerDes operates at an industry-leading **224 Gbps per lane** (using 100 Gbaud PAM4 signalling), enabling the total 1.8TB/s of bandwidth per GPU.

**Broadcom: The rising giant**

Broadcom has followed Nvidia's roadmap to introduce scale-up switches (Tomahawk Ultra) for the rising clustering demand. Broadcom's has also been working with Google on their proprietary scale-up setup, but only at XPU side, providing custom design and serdes IP to Google's TPU.

- **Approach:** Modifying its existing scale-out product (Tomahawk 5) to meet the scale-up requirement, which requires low-latency and even more memory sharing mechanism.
- **Switch Product:** Tomahawk Ultra, 51.2Tpbs capacity, launched in July 2025.
- **Protocol: Scale-up Ethernet (SUE),** which is a modified Ethernet standard suitable for AI computing with acceptable latency.
- **GPU Fabric Size:** A single Tomahawk Ultra fabric can support up to 1,024 XPUs cluster at lower interconnect speed, or 64 XPUs at higher interconnect speed.
- **Processor SerDes:** Tomahawk ultra operates at **112 Gbps per lane,** enabling the total 51.2Tbps/s of bandwidth per switch.

**Google: The switchless pioneer**

Google's approach has been to eliminate the separate network switch entirely, integrating its function directly into the AI accelerator.

- **Approach:** A "switchless" design where the TPU chip itself is the network. This is achieved by creating a direct-connect mesh topology.

- **Switch Product:** There is no separate switch. The high-speed interconnect ports are integrated directly onto the TPU silicon.
- **Protocol:** A custom Inter-Chip Interconnect (ICI) protocol.
- **GPU Fabric Size:** Google's TPU pods have scaled to thousands of chips in a single fabric (e.g., 9,216 chips in a TPU v7 pod).
- **Torus vs. Switched Fabric Limitations:** Google's torus interconnect, where each chip connects directly to its neighbours (north, south, east, west), is highly efficient for workloads with predictable, localised communication. However, it is not as scalable or flexible as a switched Clos network for certain modern AI models.
  - ⇒ **Detailed Explanation:** In a torus, communication between distant chips must make multiple "hops" through intermediate chips, increasing latency with every hop. This architecture struggles significantly with Mixture-of-Experts (MoE) models, which require sparse, unpredictable, all-to-all communication patterns. A switched fabric like NVSwitch provides a direct, non-blocking, single-hop path between ***any*** two processors, making it far superior for MoE. Due to this limitation, it is widely believed that Google is migrating towards a hybrid or fully switched fabric for future TPU generations to better handle these critical workloads.

**Amazon (AWS): Custom silicon for the Cloud**

Like Google, AWS develops custom interconnects for its own AI accelerators to optimise performance within its cloud environment.

- **Approach:** Custom-designed, high-performance link for AWS Trainium accelerators.
- **Switch Product:** Leverages switches from Broadcom, Marvell and Astera labs. Amazon has been using retimers from Astera labs to build their scale-up fabrics.
- **Protocol:** A proprietary AWS protocol. **NeuronLink v2**. This is an ultra-high-speed, non-blocking interconnect fabric within a single server node.
- **Fabric Size:** NeuronLink connects up to 16 Trainium accelerators within a single Trn1n instance in a fully non-blocking topology. This is a pure scale-up mesh ***within the node***, which is then connected to a massive scale-out EFA fabric for multi-node (UltraCluster) training. Unlike a torus, it does not rely on hop-by-hop communication within the 16-chip node.

**AMD & Astera Labs (UALink): Driving the Open Standard**

Companies like AMD and Astera Labs or others in UALink, propose an open-standard systems. Unlike others, they do not build a proprietary full scale-up systems but provide critical XPUs, networking switches or retimers that enables a scale-up system.

- **Approach:** Create an open, industry-standard alternative to NVLink.
- **Switch Product:** AMD does not produce the switch. Astera Labs is aiming to launch its scale-up switch in 2H26, and its current products are Smart DSP Retimers (like the Aries series) for PCIe, CXL, and by extension, UALink. These chips regenerate and re-time the high-speed electrical signals, allowing them to travel over longer distances (eg, across a server backplane or through cables to another chassis). Without retimers, building a large 1,024-accelerator UALink fabric would be physically impossible.
- **Protocol:** UALink 1.0 (with 1.1 specifications emerging).
- **GPU Fabric Size:** The specification is designed to connect up to 1,024 accelerators in a single, coherent memory domain.
- **Processor SerDes:** UALink leverages the open PCIe/CXL physical layer, initially targeting PCIe Gen6 speeds (~128 Gbps per lane PAM4).

**Huawei: Building a self-reliant stack**

Huawei has developed its own vertically integrated stack for the Chinese market, including a proprietary scale-up interconnect.

- **Approach:** A self-reliant, full-stack solution centred on its Ascend AI processors.
- **Switch Product:** A custom interconnect switch that is part of its CloudMatrix 3.0 architecture.
- **Protocol:** The proprietary Huawei Collective Communication Service (HCCS).

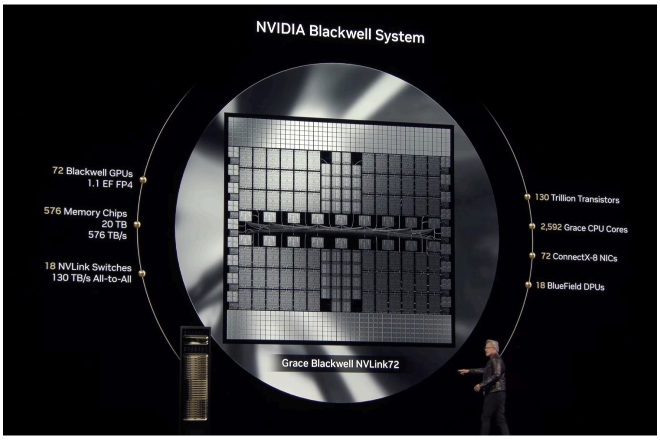
- **Fabric Size:** The Atlas 900 SuperCluster creates a single scale-up domain connecting 384 Ascend 910B AI processors.

## Figure 27 - High-speed SerDes overview

| Serdes I/O vendors | Current Serdes Speed | Current Switch capacity | Current 224G serdes status | Possible Scale-up protocol | Possible Scale-up switch: GPU ratio |
|---|---|---|---|---|---|
| Broadcom | 224Gbps | Tomahawk 6 102.4Tbps | Ready | SUE | 12:64 |
| Nvidia | 224Gbps/112Gbps | NVSwtich 5 28.8Tbps/ Spectrum X 51.2Tbps | Ready | NVLink | 18:72 |
| Marvell | 112Gbps | Teralynx 10 51.2Tbps | Switch In development | NVLink fusion/UALink | 18:72 |
| Mediatek | 112Gbps | Nephos 6.4Tbps/12.8Tbps (multi-die) | Switch In development | NVLink fusion/UALink | 18:72 |
| Astera Labs (Alchip/Synopsis) | NA | NA | IP | UALink/NVLink fusion | 12:64 |
| Qualcomm (Alphawave) | NA | NA | IP | UALink | NA |

Source: Macquarie Research, Nov 2025

## Figure 28 - NVL72 supercomputer rack system




Source: Nvidia, Nov 2025

## Figure 29 - Nvidia's scale-up hardware

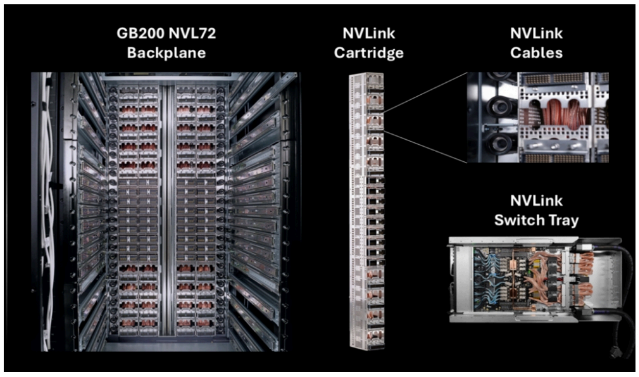
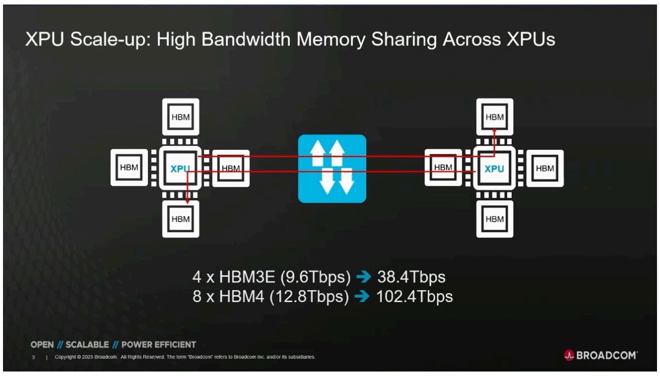



Figure 3. The NVIDIA GB200 NVL72 rack, with four NVLink cartridges housing over 5,000 energy efficient coaxial copper cables, enables each GPU to communicate with every other GPU 36x faster than state-of-the-art Ethernet standards

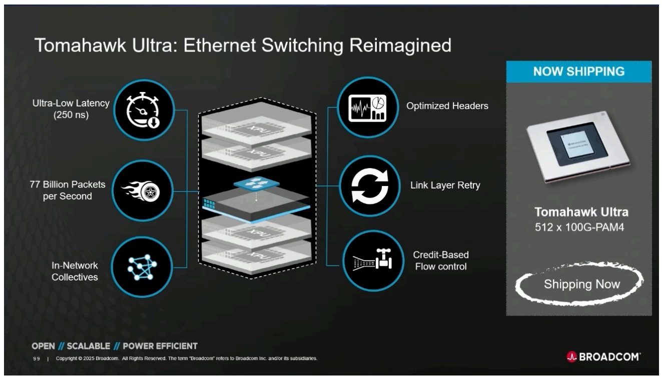
Source: Nvidia, Nov 2025

## Figure 30 - Broadcom'x XPU scale-up



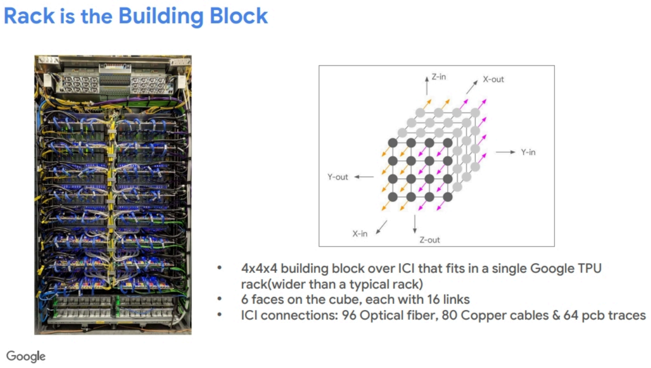

Source: Company, Nov 2025

## Figure 31 - Broadcom's scale-up switch - Tomahawk Ultra




Source: Broadcom, Nov 2025

**Figure 32 - Google's scale-up hardware**




*Source: Google, Nov 2025*

**Figure 33 - Huawei's scale-up hardware**


*Source: Huawei, Nov 2025*

# Interconnect technology: Transceivers

The introduction of additional layers underpinning the formation of AI clusters has a multiplier effect on demand for components such as high-speed transceivers. We forecast a 70% transceiver market revenue CAGR over 2024-27E, assuming 43.6mn 800G units and 13.3mn 1.6T units shipped in 2026E. Noting current shortages of PCB / CCL, we believe the next area of short supply will be transceiver-related components, particularly the light source. We expect further penetration of SiPh solutions, reaching 50% in 2026 for 800G transceivers, with Innolight and LandMark as the key beneficiaries. We also expect spill-out effects to allow second-tier light source providers to enter the market with qualification through Japanese laser providers and Taiwanese PD maker, GCS (4991 TT, NR).

**Transceiver trend**

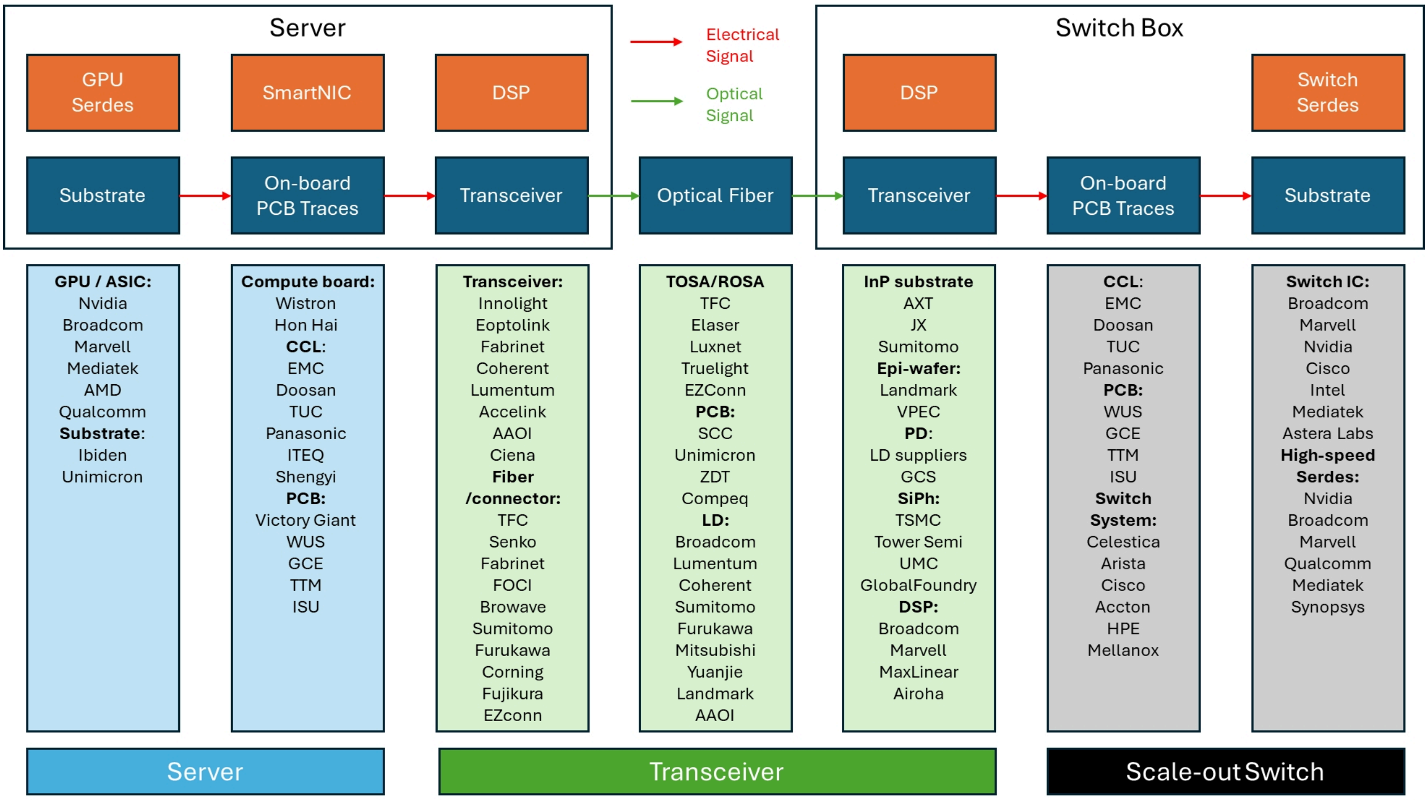
Nvidia has been actively adopting optical interconnect solutions for ITS scale-out networking since H100. We note that transceiver speed moves in line with Connect-X NIC speed, as the underlying speed for the whole AI network. We are seeing Nvidia increase the cadence of its networking speed upgrades, from three years for a doubling to 800G to just one year for the doubling to 1.6T, spurring rapid demand for optical transceiver, as it is the mainstream solution for high-speed interconnect.

**Transceivers explained**

Transceivers are a key component enabling longer-distance / higher-data rate transmissions. If we limit the power of the electrical signal, the signal can only travel a limited distance. At higher data rates, signal loss becomes more severe. We can convert electrical signals to optical signals with a reasonable conversion power cost, which enables much longer transmission distance, which is where transceivers come in, translating electrical signals to optical signals and vice versa.

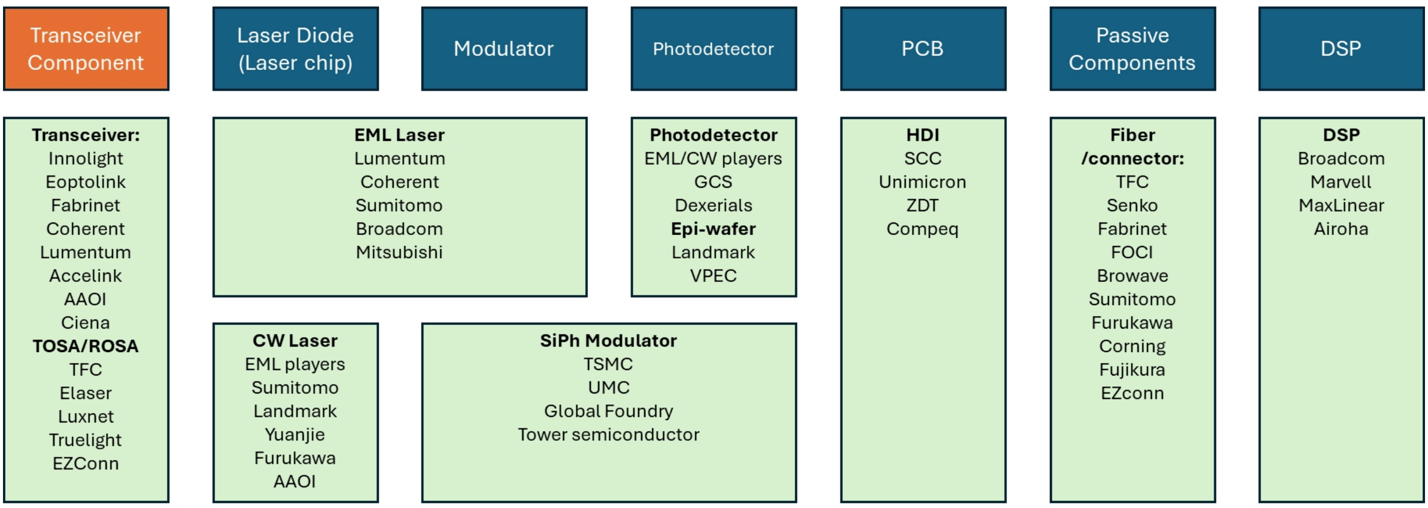
Traditionally, transceivers were only used in the spine or data-centre interconnect layer for ultra high-speed applications. With AI, we see the leaf side also adopting high-speed transceivers. In addition, transceivers benefit from the density of connections in AI networking. Due to its non-blocking / lossless requirements, AI network density is much higher than that of a traditional Ethernet, creating an even better demand outlook for transceiver.

## Figure 34 - Scale-out Opportunity for Optical Communication (public companies)




*Source: Bloomberg, Macquarie Research, Nov 2025*

## Figure 35 - Transceiver supply chain overview




*Source: Macquarie Research, Nov 2025*

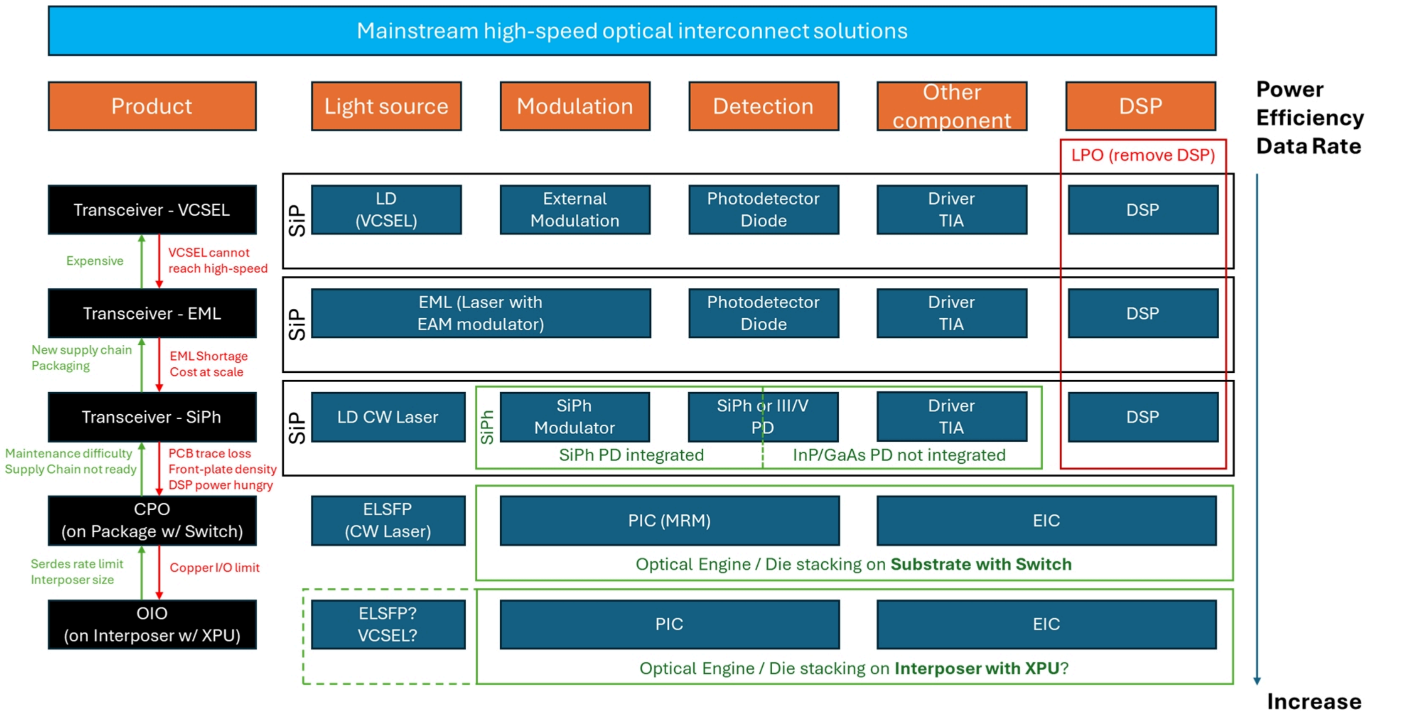
## Figure 36 - Nvidia's interconnect technology overview

| GPU | | Scale-up (In-rack) | Scale-out (Between-rack) | |
|---|---|---|---|---|
| H100 (2022) | Chip | NV Switch 4 (900 GB/s) | CX7 – 400G | Infiniband / Spectrum4 51.2T |
| | Interconnect | PCB trace | Layer 1 Transceiver 400G VCSEL | Layer 2/3 Transceiver 800G EML,SiPh |
| GB200 (2024) | Chip | NV Switch 5 (1.8TB/s) | CX7 – 400G | Infiniband / Spectrum4 51.2T |
| | Interconnect | DAC (Copper) | Layer 1 Transceiver 400G VCSEL | Layer 2/3 Transceiver 800G EML,SiPh |
| GB300 (2025) | Chip | NV Switch 6 (3.6TB/s) | CX8 – 800G | Infiniband / Spectrum4 51.2T |
| | Interconnect | DAC (Copper) | Layer 1 Transceiver 800G EML, SiPh | Layer 2/3 Transceiver 800G EML,SiPh |
| Rubin (2026) | Chip | NV Switch 7 (3.6TB/s) | CX9 – 1.6T | Infiniband / Spectrum 6 102.4T |
| | Interconnect | DAC (Copper) | Layer 1 Transceiver 1.6T EML,SiPh /CPO? | Layer 2/3 Transceiver 1.6T EML,SiPh? CPO? |
| Rubin Ultra (2027) | Chip | NV Switch 8? | CX10? | Infiniband / Spectrum 7 204T |
| | Interconnect | PCB Backplane? AEC? | Transceiver? CPO? | Transceiver? CPO? |

Increased pace

*Source: Nvidia, Macquarie Research, Nov 2025*

## Figure 37 - High-speed interconnect solutions overview




*Source: Macquarie Research, Nov 2025*

# CPO technology

The first wave of CPO beneficiaries will be testing equipment providers

We anticipate some level of real adoption in 2027E, with a possible ramp in 2028E

We are taking a conservative view on the timing of Co-Packaged Optics (CPO) adoption, as our channel checks suggest that the manufacturing process for CPO has yet to be finalised. We think the first wave of CPO beneficiaries will be testing equipment providers, before any real beneficiaries at the production level, as CPO manufacturing still face multiple challenges and testing is a major bottleneck. We anticipate some level of real adoption in 2027E, with a possible ramp in 2028E, as networking bandwidth continues to grow.

**Q: What is CPO?**

CPO is to package optical engines (which are smaller chiplets functioned as transceiver) directly next to the switch chip. It will provide power saving and signal integrity advantages at high data rate. We have laid out mainstream high-speed optical communication road maps in fig, suggesting a clear typology for different solutions. Notice that when a new solution is introduced, it does not wipe-out the whole market, similar as OLED to LCD market.

**Q: What are the arguments for CPO use? How distant is mass adoption?**

Put simply, CPO can boost signal integrity (Figure 38) and power efficiency (Figure 39). The major benefits offered by CPO solutions are power efficiency, data rate (trace loss) and scalability (front-plate density). CPO reduces copper traces, which result in lower trace loss, enabling even higher data rates and better power efficiency. Its compact design also allows more ports to be placed inside the switch box, connecting more GPUs than ever. TSMC and Nvidia have both unveiled fascinating CPO solutions, and the industry seems to be hyped on some early winners in the supply chain. Nonetheless, we have yet to see mass adoption among end-customers.

No CPO solution finalised as yet; needs a determined supplier before real mass production starts.

**Q: What are the hurdles to mass adoption?**

On the demand side, we observe some major pushback from CSP customers, either there's some worries on the maintainability / serviceability of CPO solution, concerns on supply chain concentration, and just different technology directions that are not compatible with CPO solution. For example, Google's Optical circuit switch is directly leverage light signals as input, making a CPO switch incompatible with its proprietary solution. The number of CPO suppliers is limited to only Broadcom, Nvidia and Marvell, a concentration that is the source of significant concern around CPO as a solution.

**Q: Why emphasise testing and not production?**

Testing process are not yet finalised. We have not found evidence of any large orders for CPO equipment in the market as yet, suggesting high-volume CPO production remains remote for now. CPO is a semiconductor solution, with no solution finalised as yet. It requires a determined supplier before entering real mass production. Our observation from discussions at industry forums such as OCP Asia and Semicon suggest that foundries are still trying to finalise the supply chain, as CPO players are talking about testing "frameworks", indicating that is still an early stage.

The main reason that CPO has not yet been adopted is that CPO is a new technology and the supply chain is not ready for mass production. The gap between electrical testing and optical testing is huge (Figure 40), leading to production challenges, including warpage. While we note some standardisation of testing phases thanks to efforts from TSMC and its customers (Figure 41), we believe it is an ongoing effort; we expect some major beneficiary from testing to emerge in 2026E, but we remain conservative as to the timeline for mass CPO production.

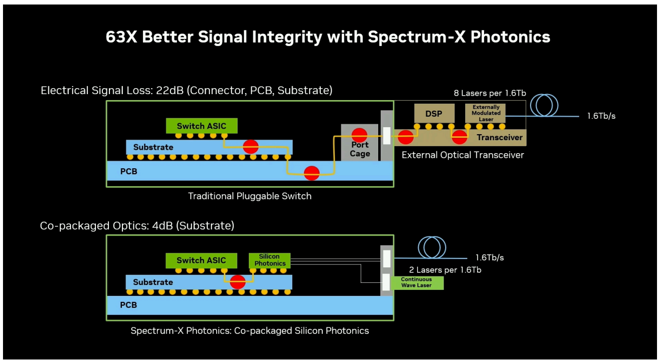
**Q: What role will transceivers play going forward?**

We also strongly believe transceiver demand will sustain, as there are still growing market opportunity ahead for transceivers.

- **Reluctant CSPs**: Google, for example, is refusing on using CPO, as its proprietary Optical Circuit Switch is inherently not compatible with CPO switches. Others CSPs are somewhat reluctant due to the

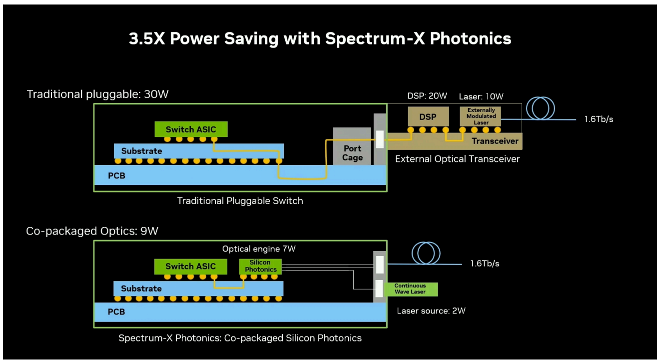
- **Scale-across:** As clusters become too large for 1 single site to handle, it is necessary to use ZR transceiver (super-long-distance transceivers) for data centres on different sites to be connected as 1 cluster.
- **China domestic interconnect demand**:
  ⇒ CPO switch requires ultra-high-speed switch silicon and advanced packaging technology.
  ⇒ Chinese foundries are only capable of legacy nodes, which are not suitable for manufacuring ultra-high-speed switch.
  ⇒ As silicon upgrade is very limited, a system-upgrade approach is more likely for China.
  ⇒ A major example will be Huawei's CloudMatrix 384, which using a massive amount of transceivers to enable a comprehensive AI system.

## Figure 38 - CPO to produce better signal integrity



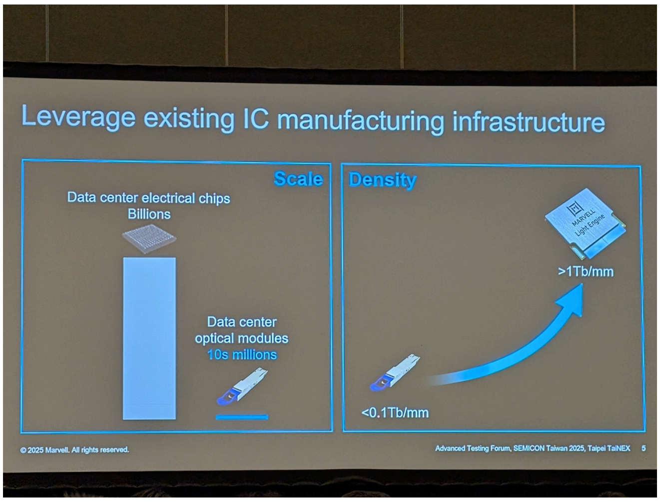

*Source: Nvidia, Nov 2025*

## Figure 39 - CPO as a power efficient interconnect solution



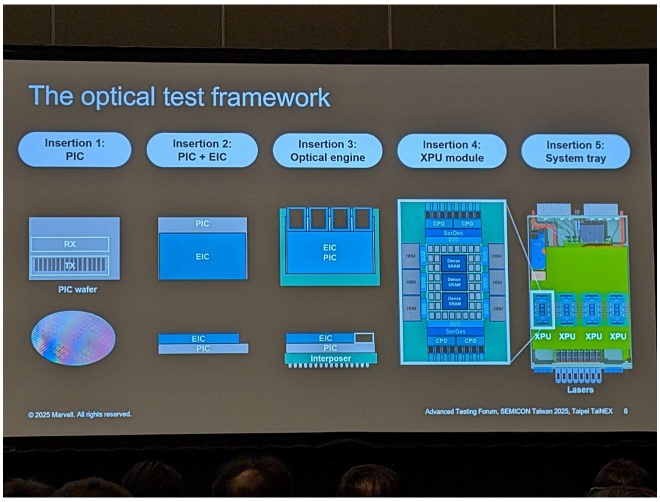

*Source: Nvidia, Nov 2025*

## Figure 40 - Optical testing still needs scaling




*Source: Marvell, OCP Asia 2025, Nov 2025*

## Figure 41 - Optical testing framework




*Source: Marvell, OCP Asia 2025, Nov 2025*

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### Recommendation definitions

**Macquarie – Asia and USA**
Outperform – expected return >10%
Neutral – expected return from -10% to +10%
Underperform – expected return <-10%

**Macquarie – Australia/New Zealand**
Outperform – expected return >10%
Neutral – expected return from 0% to 10%
Underperform – expected return <0%

During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions.

**Recommendations** – 12 months
**12-month target** – Expected share price in 12 months
**Valuation** – The company's estimated fair value share price based on the disclosed valuation methodology
**Note**: Quant recommendations may differ from Fundamental Analyst recommendations

### Volatility index definition

This is calculated from the volatility of historical price movements.

**Very high** – highest risk – Stock should be expected to move up or down 60–100% in a year – investors should be aware this stock is highly speculative.

**High** – stock should be expected to move up or down at least 40–60% in a year – investors should be aware this stock could be speculative.

**Medium** – stock should be expected to move up or down at least 25–40% in a year.

**Low** – stock should be expected to move up or down at least 15–25% in a year.
* Applicable to select stocks in Asia/Australia/NZ

**Note:** expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk

### Financial definitions

All "Adjusted" data items have had the following adjustments made:
**Added back:** goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense
Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests

**EPS** = adjusted net profit / efpowa*
**ROA** = adjusted ebit / average total assets
**ROA Banks/Insurance** = adjusted net profit /average total assets
**ROE** = adjusted net profit / average shareholders funds
**Gross cashflow** = adjusted net profit + depreciation
*equivalent fully paid ordinary weighted average number of shares

All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

### Recommendation proportions for quarter ending 30 Sept 2025

| | AU/NZ | Asia | USA | |
|---|---|---|---|---|
| **Outperform** | 51.96% | 67.82% | 67.16% | (for global coverage by Macquarie, 1.45% of stocks followed are investment banking clients) |
| **Neutral** | 38.79% | 18.36% | 32.84% | (for global coverage by Macquarie, 1.75% of stocks followed are investment banking clients) |
| **Underperform** | 9.25% | 13.83% | 0.00% | (for global coverage by Macquarie, 0.00% of stocks followed are investment banking clients) |

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